Human Vision and Image Features

Multimedia Retrieval · Appendix E

Multimedia Retrieval - 2026 - Uni Basel

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E - Human Vision and Image Features

\#image-features keeps the classical image features short. Human vision gets two paragraphs: three cone types, metamerism, and an illusion that shows that perception is not measurement. Lab and HSV appear as a choice of color space, without their conversion formulas. Gabor filters appear as a concept, without the computation behind them. This chapter supplies what the main text leaves out, for readers who want to know where the three numbers per pixel come from and how the features built on them are computed.

The details matter when a feature behaves unexpectedly: when two colors that look alike lie far apart in RGB, when a color moment computed in a single pass loses its precision, or when a filter bank is faster in Fourier space than as a convolution.

Where this is used

- \#image-features-color keeps three cone types, metamerism, and the checker-shadow illusion from The Human Visual System, and it uses Lab and HSV, whose formulas are in Color Systems.
- \#image-features-edges-texture-shape introduces Gabor filters as a concept; Extended Image Features computes them in Fourier space and adds gradient moments and Laws' texture energy.
- \#image-features-cnn builds on the receptive fields and the orientation-selective cells of the visual cortex.
- Distance and Similarity Measures treats the quadratic distance, which Extended Image Features turns into a Euclidean distance.

Prerequisites

- Color histograms, color moments, and the quadratic distance: \#image-features-color.
- Gradients and Gabor filters: \#image-features-edges-texture-shape.
- For the last section, eigenvectors of a symmetric matrix and the discrete Fourier transform.

The Human Visual System follows the light from a surface through the eye and the retina into the visual cortex. Color Systems turns the three cone signals into the color spaces that image features compute with, CIE XYZ, xyY, sRGB, Lab, HSL, and HSV, and follows one color through all of them. Extended Image Features shows how moments are computed in one pass, how the quadratic distance becomes a Euclidean distance, two further texture features, and how Gabor filters work in Fourier space.

The Human Visual System

Where this is used

\#image-features-color keeps the core of this section: three cone types, metamerism, and the checker-shadow illusion, which shows that perception is not measurement. \#image-features-cnn builds on the receptive fields and the orientation-selective cells of the visual cortex described below, because they inspired convolutional networks. The invariances that Automated Feature Extraction asks of perceptual features, such as stability under a change of light, are invariances the visual system already has. Color Systems turns the three cone signals into the color spaces that image features compute with.

This section follows the path of light from a surface to the cortex. We start with the physical signal, then look at how the eye samples it, how fine these samples are compared with other species, and how the retina and the brain process them.

Light, Illumination, and Reflectance

Every image starts with light that falls on a surface and leaves it toward a viewer. Two quantities describe this interaction. The **illumination** $l(x, y, z)$ is the amount of light that arrives at a point in space. We measure it in lux, a unit that weights the power at each wavelength by the sensitivity of the human eye. The **reflectance** $r(x, y, z)$ is the fraction of that light that the surface at this point sends back. It lies between 0 for a surface that absorbs all light and 1 for a surface that reflects all of it.

Figure 0.1 shows both sides of the interaction. On the left, light from a source with color $R_s G_s B_s$ hits a small surface fragment whose material has the reflectance $R_m G_m B_m$. Part of the light is reflected as in a mirror, at the same angle θ to the surface normal at which it arrived. The rest is scattered diffusely in all directions. The viewer sees only the light that leaves the fragment toward the eye. A small change in the direction, the intensity, or the color of the light source therefore changes how the same surface looks.

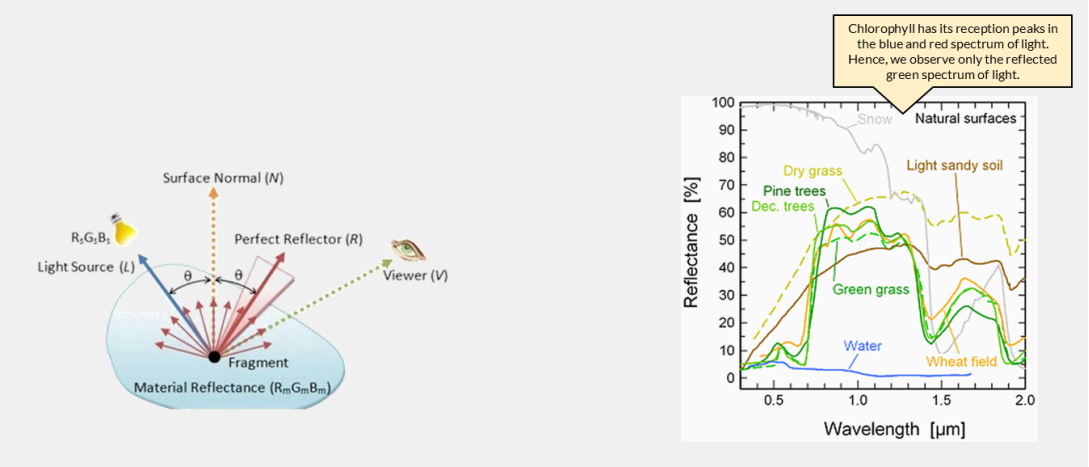


Figure 0.1: Light-surface interaction (left) and spectral reflectance of natural surfaces (right).

The right side of the figure shows the reflectance of natural surfaces for wavelengths from 0.4 to 2.0 micrometers. Snow reflects about 80 to 90 percent of the visible light. Water reflects only a few percent at all wavelengths. Vegetation reflects little visible light, has a small peak in the green near 550 nm, and reflects strongly in the near infrared above 700 nm, which we cannot see. Leaves look green because chlorophyll absorbs mainly blue light (around 430 to 455 nm) and red light (around 640 to 665 nm). Green light is absorbed less, and the cell structures of the leaf scatter it back toward the viewer.

These curves show that the physical signal is a rich, continuous function of the wavelength. Human vision and camera sensors reduce it to three numbers per point. Color Systems describes this reduction in detail.

Key Formula: Image Formation

$$f(x, y, z) = l(x, y, z) \cdot r(x, y, z), \qquad 0 < l(x, y, z) < \infty, \quad 0 < r(x, y, z) < 1 \tag{0.1}$$

In the simplest model, the light f that a point sends toward the viewer is the product of illumination and reflectance. The same product holds for each wavelength separately: $f(\lambda) = l(\lambda) \cdot r(\lambda)$.

The two factors vary on very different scales. Direct sunlight delivers 32,000 to 100,000 lux, an overcast day about 1,000 lux, office lighting 320 to 500 lux, and a full moon on a clear night only 0.05 to 0.3 lux. Reflectance, in contrast, spans from about 1 percent for a black fabric to about 90 percent for fresh snow. The illumination thus changes by a factor of 100,000 and more between day and night, the reflectance only by a factor of about one hundred.

Example

Consider a piece of black velvet with $r = 0.01$ in direct sunlight with $l = 100,000$ lux, and a sheet of white paper with $r = 0.8$ under office lighting with $l = 500$ lux:

$$f_{\text{velvet}} = 100,000 \cdot 0.01 = 1,000, \qquad f_{\text{paper}} = 500 \cdot 0.8 = 400 \tag{0.2}$$

The velvet sends 2.5 times more light into the eye than the paper. Still, we see the velvet as black and the paper as white. The visual system does not report the product f. It estimates the reflectance r and discounts the illumination, an ability called **lightness constancy**. A feature extractor that only sees pixel values, which are proportional to f, would call the velvet the brighter surface.

Lightness constancy also explains the checker-shadow illusion in \#image-features-color: two squares that send the same light to the eye look different, because the visual system attributes the darkness of one of them to a shadow. For feature design, the formula suggests a remedy. Two neighboring points under the same light have the ratio $f_1/f_2 = (l \cdot r_1)/(l \cdot r_2) = r_1/r_2$, so the illumination cancels. Features that compare neighboring pixels, such as edges and gradients, are therefore more robust to lighting than features that use absolute brightness. The retina, as we will see below, works with exactly such comparisons.

The Eye and Its Photoreceptors

The eye converts light into electrochemical signals that the brain can interpret. Figure 0.2 shows its structure. Light enters through the cornea, passes the pupil, and goes through the lens. Together, the cornea and the lens focus the image of an object onto the retina at the back of the eye. The lens changes its shape to focus on objects at different distances, and the iris changes the size of the pupil to adapt to the light level. A camera does the same with its focus and its aperture. The image on the retina is upside down, and the brain interprets it correctly.

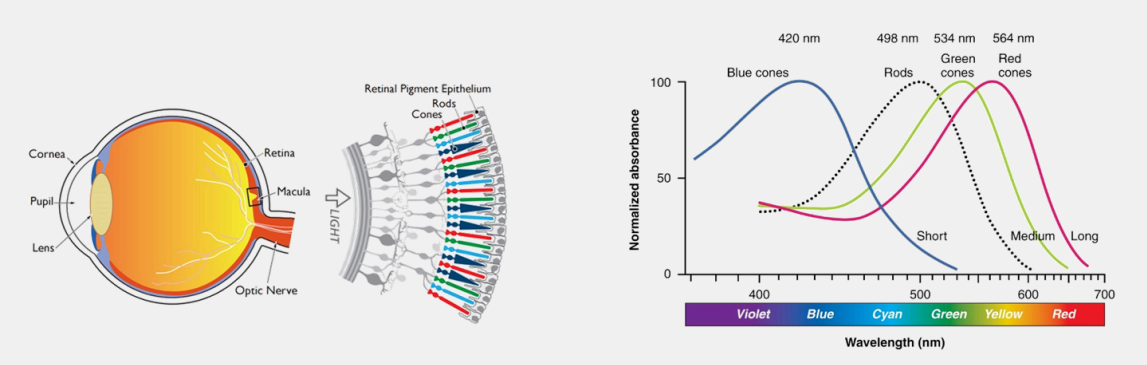


Figure 0.2: The human eye, the layers of the retina, and the normalized absorbance of the three cone types and the rods.

The magnified part of the figure shows a surprising detail of the retina. Light passes through several layers of nerve cells before it reaches the photoreceptors, which point away from the light toward the pigment epithelium. There are two kinds of photoreceptors, rods and cones, and they serve different purposes.

Cones provide sharp, detailed, and colored vision in daylight. They are concentrated in the macula and in its center, the fovea, which is the part of the retina we point at whatever we look at. There are three types of cones with different sensitivity curves, shown on the right of the figure:

- **S cones** (short wavelengths) peak near 420 nm, in the violet-blue.
- **M cones** (medium wavelengths) peak near 534 nm, in the green.
- **L cones** (long wavelengths) peak near 564 nm, in the yellow-green.

The common names “blue”, “green”, and “red” cones are convenient but misleading for the L cones. Their peak is not in the red. They earn their name because their sensitivity reaches furthest into the long wavelengths, so red light excites them much more than the M cones. The M and L curves overlap strongly. A single wavelength therefore excites two or three cone types at once, and the brain derives color from the comparison of their responses.

Rods dominate the periphery of the retina and work best in low light. They are essential for night vision and for noticing movement at the edge of the visual field. All rods share one pigment with a peak near 498 nm. With only one type of receptor, the rods cannot distinguish colors, which is why we see shades of gray at night.

The numbers of photoreceptors are large and vary between people. Curcio et al. (1990) counted an average of 4.6 million cones (4.1 to 5.3 million) and 92 million rods (78 to 107 million) per retina, with up to 199,000 cones per square millimeter in the center of the fovea. Many textbooks quote rounded values of 6 million cones and 120 million rods; the measured averages are lower.

The three cone types are far from equally frequent. S cones make up only about 7 percent of the cones near the fovea (Curcio et al., 1991). They are irregularly spaced, missing entirely in a small zone of about 0.35 degrees at the center of the fovea, and most dense in a ring just outside it. The ratio of L to M cones varies even more. In eight men with normal color vision, Hofer et al. (2005) measured ratios from 1.1:1 to 16.5:1. Fixed percentages for the three cone types should therefore be read as one example, not as a constant of human vision.

These variations do not cause color blindness: all the men in the study above saw colors normally. Color blindness arises when one cone type is missing or when its pigment is shifted to a different wavelength. The most common forms affect the L or the M cones and make red and green hard to tell apart.

Visual Acuity

Visual acuity measures how fine a detail the eye can resolve. The standard test is the Snellen chart: rows of letters of decreasing size, read from a distance of 20 feet (6 meters). A result of 20/20, written 6/6 in metric units, means that the eye resolves the critical details of a letter, such as the gap in a “C” or the width of a stroke, when they cover a visual angle of 1 arcminute (1/60 of a degree). The whole letter is five times larger and covers 5 arcminutes. At 20 feet, 1 arcminute corresponds to

$$20 \ \text{ft} \cdot \tan\left(\frac{1}{60}^\circ\right) = 6.096 \ \text{m} \cdot 0.000291 = 1.77 \ \text{mm}, \tag{0.3}$$

so a person with 20/20 vision separates two contours about 1.77 mm apart at that distance, and the letters of that row are about 8.9 mm tall. A result of 20/40 means that the person must stand at 20 feet to read what a person with normal vision reads at 40 feet. The details must be twice as large, 2 arcminutes instead of 1.

Clinical practice also expresses acuity as a decimal number, the reciprocal of the minimum angle of resolution (MAR) in arcminutes:

$$V = \frac{1}{\text{MAR}} \tag{0.4}$$

A result of 20/20 gives $V = 1.0$, and 20/40 gives $\text{MAR} = 2$ and $V = 0.5$. Vision science often uses a third unit, the number of grating cycles per degree that the eye can still resolve. One cycle consists of one dark and one bright bar. If each bar covers 1 arcminute, one degree holds 30 cycles, so 20/20 corresponds to 30 cycles per degree, and 20/40 to 15. In general, a Snellen fraction $20/D$ corresponds to $600/D$ cycles per degree. We use this conversion in the comparison of species below.

The value 20/20 is the lower limit of normal vision, not the average. Healthy young adults typically reach 20/16 to 20/12 (6/5 to 6/4), and the limit of the unaided human eye lies around 20/10 to 20/8, or about

60 cycles per degree. Driving licences set minimum values. In the European Union, a car driver needs a binocular acuity of at least 0.5 with both eyes, which is a MAR of 2 arcminutes or 20/40. The German screening test for a car licence is stricter and asks for 0.7, a MAR of about 1.4 arcminutes.

Vision in Other Species

Animals perceive the world very differently. Figure 0.3 shows on the left the same farmhouse scene as a human sees it and as a cat sees it: the cat's version has fewer colors and much less detail. The right side shows the four cone types of goldfish and zebrafish, which reach into the ultraviolet.

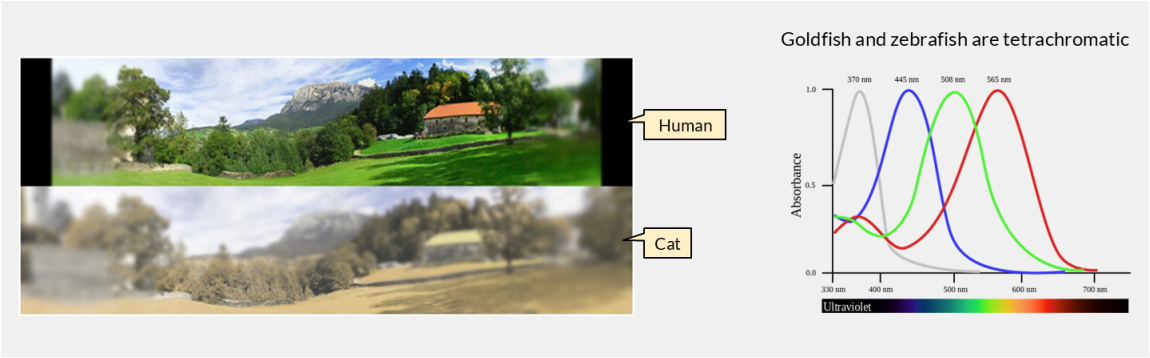


Figure 0.3: The same scene for human and cat vision (left) and the four cone types of goldfish and zebrafish (right).

The following table compares the acuity of several species. The Snellen equivalents use the conversion $600/D$ from the previous subsection, and all values are approximate.

Species	Acuity (cycles per degree)	Snellen equivalent
Wedge-tailed eagle	132 to 143	about 20/4
Human, limit of the eye	about 60	20/10
Human, normal lower limit	30	20/20
Dog	about 8	20/75
Cat	3 to 6	20/100 to 20/200
Asian elephant	about 3	about 20/200
Rat	about 1	about 20/600
Mouse	about 0.5	about 20/1200
Most insects	below 1	worse than 20/600

Cats have two cone types, with peaks near 450 nm and 550 nm, so their color vision is limited. Their acuity of 20/100 to 20/200 means that a cat must be at 6 meters to see what a human sees at 20 to 30 meters. In return, cats see in dim light with about one-sixth of the light that humans need. They have many rods and a reflective layer behind the retina, the tapetum lucidum, which sends the light that passed the photoreceptors back through them for a second chance of absorption. This layer also makes cat eyes shine when a light falls on them at night. The field of view of a cat is about 200 degrees, compared with about 180 degrees for humans, which helps with hunting but does not add detail.

Dogs are also dichromats. Their two cone types peak near 429 nm and 555 nm, and their color vision resembles red-green color blindness in humans. Their acuity of about 20/75 is a bit better than that of cats but still far below human acuity. **Asian elephants** resolve gaps of 8 to 11 arcminutes, about 20/200. Small **rodents** see much less clearly: rats resolve about 1 cycle per degree and mice about 0.5. **Insects** see with compound eyes, whose acuity is limited by the angle between neighboring ommatidia, the small units of the eye. Most insects resolve less than 1 cycle per degree, and humans resolve more than a hundred times more detail than a fruit fly.

Birds of prey are the exception. The wedge-tailed eagle resolves 132 to 143 cycles per degree (Reymond, 1985), about twice the human limit. This lets it spot prey from great distances. Claims that eagles see four to

eight times sharper than humans are not supported by measurements. Many birds also have four types of cones instead of three: they are **tetrachromats**, and the fourth cone type is sensitive to violet or ultraviolet light that humans cannot see. Goldfish and zebrafish are tetrachromats as well, with an extra cone type for ultraviolet light, useful for detecting UV light under water. The curves in the figure are schematic, with peaks near 370, 445, 508, and 565 nm. Measured peaks in zebrafish are 361, 411, 482, and 565 nm (Allison et al., 2004).

The comparison has a practical consequence for retrieval. Color spaces such as RGB are built for the three human cone types. They describe what a human sees, not what a bird, a fish, or a dog sees, and they are the right choice exactly because a human judges the search results.

Processing in the Retina

The retina does more than capture light. It already processes the signal, and this processing inspired the use of convolutions in deep learning. Figure 0.4 shows the main cell types and how they respond to light.

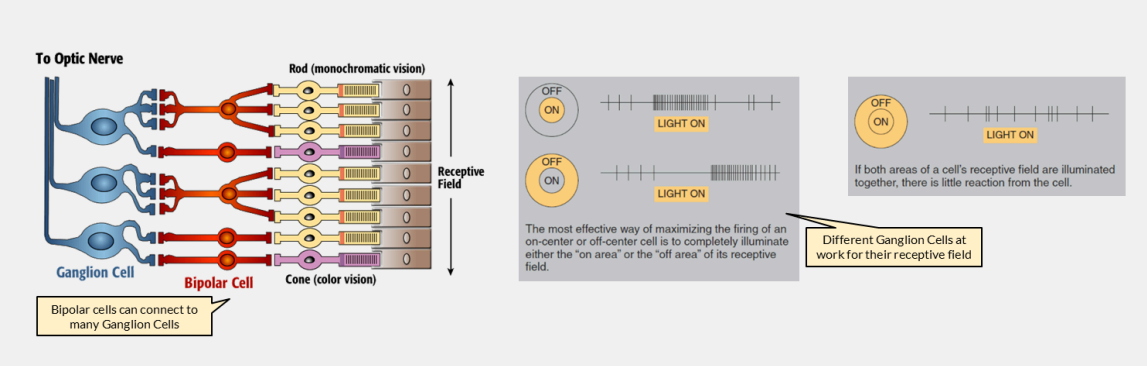


Figure 0.4: Rods, cones, bipolar cells, and ganglion cells (left), and the responses of a ganglion cell to light on the center, on the surround, and on both (right).

The photoreceptors show an unusual chemical behavior. In the dark, they release the neurotransmitter glutamate continuously. Light reduces this release. Most sensory receptors work the other way round and release more transmitter when they are stimulated. The continuous release in the dark costs energy, but it lets one signal carry changes in both directions: more light means less glutamate, less light means more.

The next stage, the **bipolar cells**, reads this signal in two opposite ways. In the primate retina, each bipolar cell collects its input from either rods or cones, not from both. **ON bipolar cells** respond to light with depolarization, which means they become more active when the light increases. **OFF bipolar cells** respond with hyperpolarization: they become less active in light and more active when the light decreases. Bipolar cells do not fire spikes. They signal with graded changes of their membrane potential.

The **ganglion cells** combine the signals of several bipolar cells and are the output neurons of the retina. Their axons form the optic nerve, and they send their signals as spikes. Each ganglion cell responds to light in a small region of the visual field, its **receptive field**. The receptive field has a center and a surround with opposite effects. An on-center ganglion cell fires strongly when light falls on the center and the surround is dark, and it is suppressed when light falls only on the surround. An off-center cell does the opposite. The right side of the figure shows the key observation: if light covers both center and surround, the cell barely reacts. Ganglion cells therefore do not measure brightness. They detect local contrast, such as a bright spot, a dark spot, or an edge that passes through the receptive field.

The retina also compresses the signal. About 97 million rods and cones feed only 0.7 to 1.5 million ganglion cells (Curcio & Allen, 1990). In the periphery, many rods converge onto one bipolar cell, which pools light and increases sensitivity at the cost of detail. In the fovea, convergence is low, so that each cone keeps an almost private line to the brain and the image stays sharp. A bipolar cell can also feed several ganglion cells, as the left side of the figure shows.

The center-surround receptive field is easy to write as a computation. A ganglion cell forms a weighted sum of the light intensities I_{ij} in its receptive field, with positive weights w_{ij} in the center and negative weights in the surround:

$$g = \sum_{i,j} w_{ij}\, I_{ij} \tag{0.5}$$

This is exactly the operation of a convolution kernel in image processing.

Example

Take a 3×3 receptive field with the weight +8 in the center and −1 for each of the eight surrounding positions. The weights sum to zero.

- Uniform light with intensity 10 everywhere: $g = 8 \cdot 10 - 8 \cdot 10 = 0$. The cell does not react to diffuse light.
- A bright spot of intensity 10 on a dark surround of intensity 2: $g = 8 \cdot 10 - 8 \cdot 2 = 64$. An on-center cell fires strongly.
- A dark spot of intensity 2 on a bright surround of intensity 10: $g = 8 \cdot 2 - 8 \cdot 10 = -64$. The on-center cell is suppressed, and an off-center cell, which computes $-g$, fires strongly.

Doubling the illumination doubles all intensities and therefore doubles g, but a uniform field still gives 0. The cell responds to the reflectance differences within its receptive field, not to the overall level of light.

Two further cell types refine this processing. **Horizontal cells** connect neighboring photoreceptors, and **amacrine cells** connect neighboring bipolar and ganglion cells. Both act as inhibitors: an active cell suppresses its neighbors. This **lateral inhibition** amplifies the difference between strong and weak signals and sharpens contrast. It also exaggerates boundaries. Near the border between a dark and a bright region, the bright side looks brighter and the dark side looks darker than they are, an illusion known as Mach bands.

After-images, the visual impressions that persist after a stimulus is gone, have a different cause. When we look at a bright pattern for a long time, the photoreceptors and the retinal neurons adapt and become less sensitive. When we then look at a uniform surface, the adapted regions respond less than their neighbors, and we see a negative image of the pattern until the adaptation fades.

From the Retina to the Visual Cortex

Figure 0.5 shows the path of the signal from the eyes to the visual cortex. The optic nerves of both eyes meet at the optic chiasm. There, the fibers from the nasal half of each retina cross to the other side, while the fibers from the temporal halves stay on their side. As a result, each side of the brain receives the opposite half of the visual field from both eyes: the left visual field is processed by the right hemisphere, and the right visual field by the left hemisphere. After the chiasm, the optic tract leads to the lateral geniculate nucleus (LGN) in the thalamus, and from there the optic radiation, labeled geniculocalcarine tract in the figure, leads to the primary visual cortex (V1) in the occipital lobe.

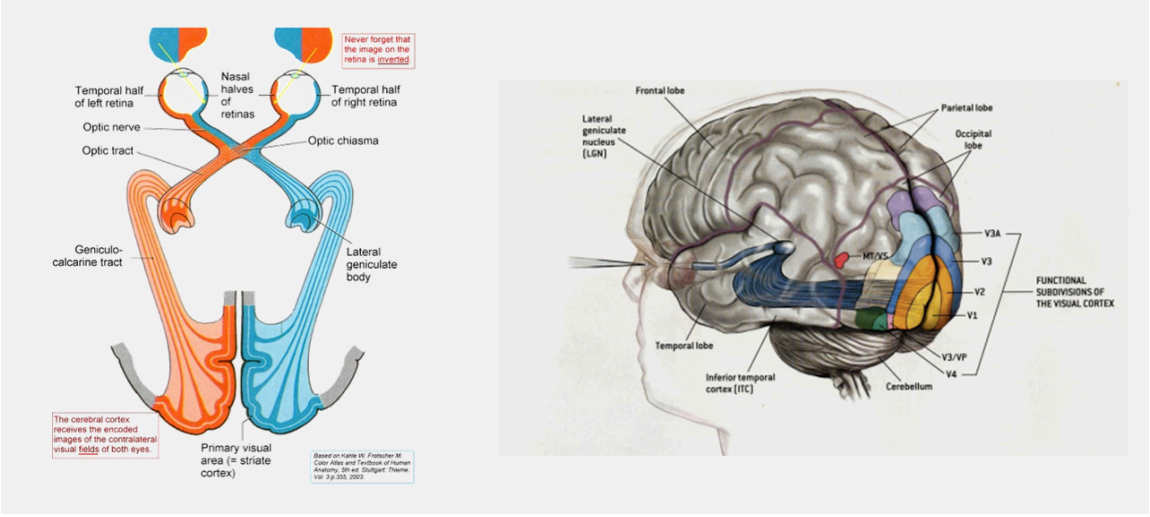


Figure 0.5: The visual pathway from the retina to the primary visual cortex (left) and the visual areas of the cortex (right).

The **lateral geniculate nucleus** is more than a relay station. Its neurons have center-surround receptive fields similar to those of retinal ganglion cells, and it receives strong feedback from the visual cortex. The LGN has six main layers. Each layer receives input from only one eye; the signals of the two eyes are combined later, in V1. Layers 1 and 2, the magnocellular layers, carry fast signals with high contrast sensitivity that are well suited for motion but carry little color. Layers 3 to 6, the parvocellular layers, carry signals for color and fine detail of form. Thin koniocellular layers between them carry the blue-yellow signals of the S cones.

The **primary visual cortex** detects edges and their orientations. Hubel and Wiesel (1962) found two kinds of neurons there. **Simple cells** respond to an edge or a bar with a particular orientation at a particular position in their receptive field. **Complex cells** respond to the same orientation anywhere within a larger receptive field, so they tolerate small shifts. Neurons in the visual cortex fire when particular patterns appear in their receptive fields. Lower levels of the system respond to simple patterns such as edges. Higher levels respond to more complex patterns: area V4 to color and form, area MT (also called V5) to motion, and regions of the inferior temporal cortex to objects and faces.

From V1, the information follows two main paths to these higher levels (Goodale & Milner, 1992):

- The **ventral stream** leads from V1 into the temporal lobe. It handles form recognition and object representation, answers the question "what is it?", and connects to long-term memory.
- The **dorsal stream** leads from V1 into the parietal lobe. It processes motion and the location of objects and coordinates eye and arm movements, for example when we reach for a cup.

The cortex does not represent all parts of the visual field equally. **Cortical magnification** means that the brain uses many more neurons for the center of the visual field than for the periphery. The fovea is a small spot on the retina, but it occupies a large part of V1. The reason is the high density of cones in the fovea and their low convergence onto ganglion cells: the cortical area devoted to a region of the visual field grows with the density of its ganglion cells (Curcio & Allen, 1990). The magnified central input lets us see fine detail, for example when we read or recognize faces.

The Visual System and Deep Networks

The human visual system and deep networks for image classification share a basic design. Both detect local features, process them in a hierarchy of layers, and learn from examples. The resemblance is not a coincidence. The simple and complex cells of Hubel and Wiesel inspired the Neocognitron (Fukushima, 1980), a network of alternating layers of "S-cells" that detect local patterns and "C-cells" that tolerate their shift. Convolutional networks keep this pattern: convolution layers play the role of the simple cells, and pooling layers the role of the complex cells. \#image-features-cnn shows that the first-layer filters of a trained network resemble the oriented edge detectors of V1.

The differences are just as important. The human system is biological. It is limited in processing speed and capacity, but it adapts well to varied visual input and learns new categories from few examples. It also relies heavily on feedback, as the connections from the cortex back to the LGN show, while a standard convolutional network passes information only forward. Deep networks are artificial. They process very large amounts of data quickly and become highly efficient at the specific tasks they were trained for, but they can fail on changes of the input that humans hardly notice.

Color Systems

Where this is used

\#image-features-color compares images with color histograms and color moments, and it shows with four boxes of sRGB colors that the choice of color space decides whether color distances make sense. It presents Lab and HSV as the main choices. This section gives the definitions and the conversion formulas behind that choice. Chapter 12 on video features removes shadows in motion detection by comparing the chromaticity of pixels, the brightness-free part of color defined below. The Human Visual System describes the cones whose responses every color system starts from.

We follow one color through the whole section: the sRGB value (255, 200, 100), a warm light orange and one of the four boxes of \#image-features-color. Step by step, we convert it to linear RGB, CIE XYZ, CIE Lab, and HSV, and at the end we compare all its coordinates with those of the same color in a shadow.

From a Spectrum to Three Numbers

Color perception is the eye's way of representing how the energy of light is distributed over the wavelengths. Figure 0.6 shows the spectrum $f(\lambda)$ of the light that reaches the eye from one observed point, together with the sensitivity curves of the three cone types and of the rods. For a surface, the spectrum is the product of the illumination and the reflectance at each wavelength, $f(\lambda) = l(\lambda) \cdot r(\lambda)$, as The Human Visual System explains.

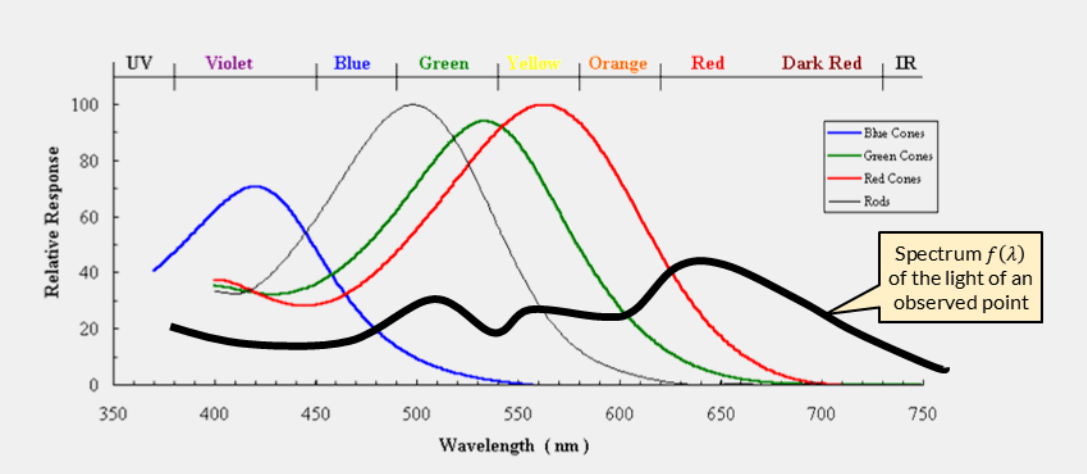


Figure 0.6: The spectrum $f(\lambda)$ of an observed point and the sensitivity curves of the cones and rods.

Each cone type responds with one number. To compute it, we multiply the spectrum with the sensitivity curve of the cone type and integrate the product over all wavelengths. With the sensitivities $\bar{l}(\lambda)$, $\bar{m}(\lambda)$, and $\bar{s}(\lambda)$ of the L, M, and S cones, the three responses are

$$L = \int f(\lambda)\,\bar{l}(\lambda)\,d\lambda, \qquad M = \int f(\lambda)\,\bar{m}(\lambda)\,d\lambda, \qquad S = \int f(\lambda)\,\bar{s}(\lambda)\,d\lambda \tag{0.6}$$

The rods would add a fourth value, but they hardly contribute to color: in daylight they are saturated, and at night the cones are silent. Color vision in daylight therefore rests on three numbers.

This reduction is only an approximation. A spectrum is a continuous function with infinitely many values, and the three integrals keep only three of them. Much information is lost, and many different spectra produce the same three responses. Such spectra look identical to us. They are called **metamers**, and we compute an example in the subsection on CIE XYZ below.

Metamerism is also what makes color displays possible. Because only three numbers matter, three primary lights suffice to reproduce most colors: a display mixes red, green, and blue light so that the mixture excites the three cone types in the same proportions as the original spectrum. The primaries do not need to match the peaks of the cones; they only need to excite the three cone types differently enough. The family of RGB systems is designed for human vision in this sense. It may not work for animals with different cone

sensitivities: a bird with a cone type for ultraviolet light sees patterns on a flower that no RGB display reproduces.

A Family of Color Systems

Besides RGB, many other color systems exist. The most common ones are:

- **CIE systems** (XYZ, xyY, Lab), defined by the International Commission on Illumination (CIE) to link the physical signal to the perception of a typical human observer.
- **RGB systems**, the prevailing systems in computing since HP and Microsoft introduced sRGB in 1996.
- **HSL and HSV**, which convert the Cartesian RGB coordinates into cylindrical coordinates for hue and saturation, with lightness or value as the third component.
- **YUV and its relatives**, which separate a brightness signal from two color-difference signals. The PAL television standard used YUV, the NTSC standard the related YIQ, and image and video compression such as JPEG and MPEG uses YCbCr. We do not discuss them further.
- **CMYK**, the system of printing. It subtracts color from a white sheet: the inks absorb light, and a blend of cyan, magenta, yellow, and black inks produces the intended color. We do not discuss it further.

Brightness, Chromaticity, Hue, and Saturation

Human vision uses three types of cones, so three components are needed to represent the full range of colors. A common way to split a color into three components is the following:

- **Brightness** is the visual perception of emitted or reflected light. It depends on the luminance of an object but is not proportional to it; the observer interprets it subjectively.
- **Chromaticity** is the objective description of a color without its luminance. It has two independent components, hue and saturation. A chromaticity diagram shows the full range of visible chromaticities, the gamut of human vision. The subset that a device or a color space can reproduce is its **color gamut**.
    - **Hue** describes how closely a color matches a pure spectral color such as red, green, or blue. In the chromaticity diagram, we draw a line from the white point through the color until it hits the curved boundary. The wavelength at this point is the **dominant wavelength** of the color. Hue is often given as an angle around the white point.
    - **Saturation** describes how concentrated the light is around a single wavelength. A pure spectral color has saturation 1; white has saturation 0. In the chromaticity diagram, saturation is the distance from the white point relative to the maximum possible distance in that direction. Color science distinguishes saturation from the related terms **chroma** and **colorfulness**, which measure the strength of a color relative to the brightness of a white or in absolute terms. Informally, the three words are often used interchangeably.

Figure 0.7 shows such a chromaticity diagram. The curved boundary, the **spectral locus**, holds the pure spectral colors from 380 nm to 700 nm. The straight line at the bottom, the **line of purples**, connects its two ends and holds colors that no single wavelength produces. The point D65 is the standard white point for average daylight, with a correlated color temperature of about 6500 K and the coordinates $(0.3127, 0.3290)$. The black curve through the interior is the **Planckian locus**. It shows the chromaticity of a black body as it heats up, from an orange-red at 1500 K through white at about 6000 K toward a bluish white as the temperature approaches infinity. A light source whose chromaticity lies close to this curve is described by its color temperature, which is how cameras and displays specify white balance.

source: https://en.wikipedia.org/wiki/Color_temperature

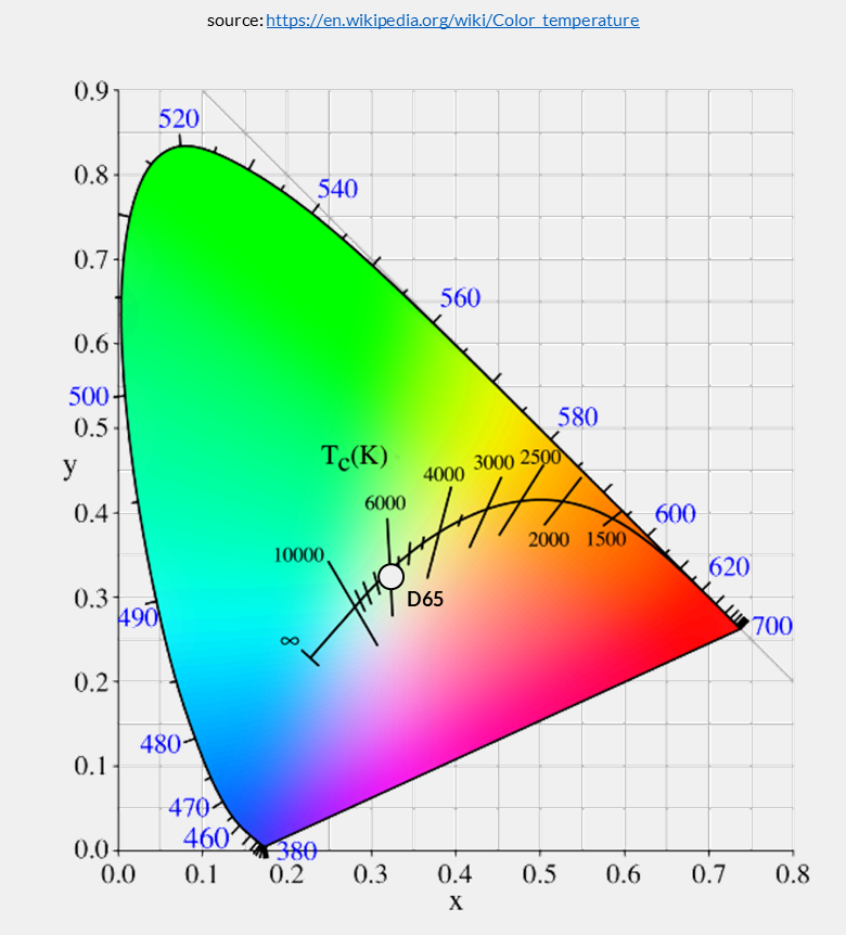


Figure 0.7: Chromaticity diagram with the Planckian locus (color temperatures in Kelvin) and the D65 white point.

The diagram is not perceptually uniform. Green lies much farther from the white point than red or blue: the distance from D65 to the spectral locus is 0.56 at 520 nm, but only 0.38 at 620 nm and 0.34 at 460 nm. This large green region is an artifact of the diagram, not a sign that humans see more shades of green. We return to this problem with CIE Lab.

CIE XYZ

The CIE defined its color spaces to represent how people see color, and the mathematics behind them is the basis of color management. In 1931, the CIE introduced the XYZ color space to model human color perception. The experiments behind it showed that people see green as brighter than red or blue when all have the same physical power; the sensitivity of the eye to brightness peaks at 555 nm. At night, the rods take over from the cones. Vision becomes nearly monochrome and much more sensitive to dim light. The rod pigment absorbs best near 498 nm, and measured through the lens of the eye, night vision is most sensitive near 507 nm.

To compute X, Y, and Z from spectral data, the CIE defined a **standard colorimetric observer** from color matching experiments. It represents the average color response of a typical person within a 2 degree field in the fovea, the central region of the retina where most cones are concentrated. A second observer for a 10 degree field followed in 1964 and is recommended for color patches larger than about 4 degrees. The standard observer is given by three **color matching functions** $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, and $\bar{z}(\lambda)$, shown in Figure 0.8. They play the role of the cone sensitivities in the integrals above.

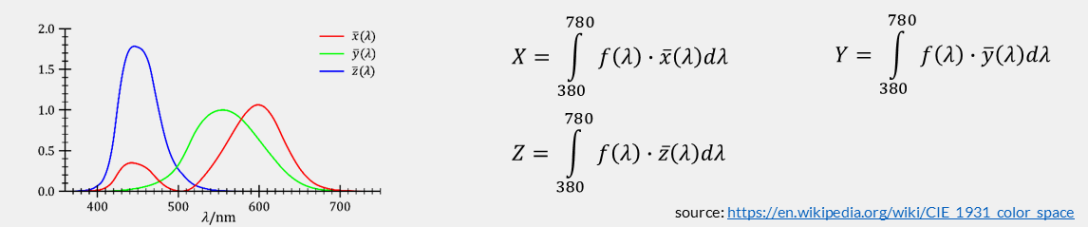


Figure 0.8: The CIE 1931 color matching functions and the integrals for X, Y, and Z.

The three components differ from the primaries of an additive or subtractive color model. Y is the **luminance**: the function $\bar{y}(\lambda)$ equals the luminous efficiency of the eye, and in terms of cones it mixes the L and M responses. Z is close to the response of the S cones. X is a mix of all three cone responses, chosen so that all three color matching functions are non-negative. The CIE chose the primaries so that Y alone carries the luminance: the primaries behind X and Z have zero luminance, so changing X or Z at a fixed Y changes the color but not its brightness. XYZ does not yet separate brightness from color, however. All three values double when the light becomes twice as bright, and the color lies only in their ratios. The xyY space of the next section makes this separation explicit.

Key Formula: CIE XYZ Tristimulus Values

$$X = k \int_{380}^{780} f(\lambda)\, \bar{x}(\lambda)\, d\lambda, \qquad Y = k \int_{380}^{780} f(\lambda)\, \bar{y}(\lambda)\, d\lambda, \qquad Z = k \int_{380}^{780} f(\lambda)\, \bar{z}(\lambda)\, d\lambda \quad (0.7)$$

Here, $f(\lambda)$ is the spectral power of the light and λ the wavelength in nanometers.

Each component is a weighted sum of the spectrum, with the color matching function as the weight. The constant k sets the scale. For a surface with reflectance $r(\lambda)$ under an illuminant $l(\lambda)$, we use $f(\lambda) = l(\lambda)\, r(\lambda)$ and $k = 100/\int l(\lambda)\, \bar{y}(\lambda)\, d\lambda$, so that a perfect white reflector has $Y = 100$ (or $Y = 1$ with 1 instead of 100). For a light source measured in watts, $k = 683$ lumen per watt gives Y in absolute units of luminance.

In practice, the color matching functions are tabulated in steps of 1 nm, and the integral becomes a sum over the samples, $X \approx k \sum_i f(\lambda_i)\, \bar{x}(\lambda_i)\, \Delta\lambda$.

Example

We compare two very different spectra. Spectrum A is flat: one unit of power per nanometer from 380 to 780 nm, 401 units in total. Spectrum B consists of three narrow lines at 450, 540, and 610 nm with powers of 59.42, 75.48, and 64.79 units, 199.7 units in total.

For spectrum A, the sums of the tabulated functions from 380 to 780 nm are $\sum \bar{x} = 106.86$, $\sum \bar{y} = 106.86$, and $\sum \bar{z} = 106.85$. We choose $k = 100/106.86 = 0.9358$ so that A has $Y = 100$, and obtain $X = 100.0$, $Y = 100.0$, $Z = 99.99$.

For spectrum B, the sums have only three terms. The color matching functions at the three wavelengths are $(\bar{x}, \bar{y}, \bar{z}) = (0.3362, 0.0380, 1.7721)$ at 450 nm, $(0.2904, 0.9540, 0.0203)$ at 540 nm, and $(1.0026, 0.5030, 0.0003)$ at 610 nm:

$$\begin{aligned} X &= 0.9358 \cdot (59.42 \cdot 0.3362 + 75.48 \cdot 0.2904 + 64.79 \cdot 1.0026) = 0.9358 \cdot (19.98 + 21.92 + 64.96) = 100.0 \\ Y &= 0.9358 \cdot (59.42 \cdot 0.0380 + 75.48 \cdot 0.9540 + 64.79 \cdot 0.5030) = 0.9358 \cdot (2.26 + 72.01 + 32.59) = 100.0 \\ Z &= 0.9358 \cdot (59.42 \cdot 1.7721 + 75.48 \cdot 0.0203 + 64.79 \cdot 0.0003) = 0.9358 \cdot (105.30 + 1.53 + 0.02) = 99.99 \end{aligned} \quad (0.8)$$

Both spectra give the same tristimulus values, so they look like the same white to the standard observer: they are metamers. Spectrum B needs only half the power of A for the same luminance, because it puts all its power where the color matching functions are large. Lamps with three narrow emission bands exploit this effect.

A key strength of CIE XYZ is its completeness. It covers every color visible to the standard observer with non-negative values. The price is that its primaries, the colors with coordinates $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, are imaginary: no physical light produces them. Many colors in the XYZ space fall outside the gamut of standard RGB spaces, because no display with three real primaries reproduces all visible colors.

CIE XYZ is widely used in colorimetry, color management, and image processing. It is the basis of the color spaces CIE Lab, CIE Luv, and CIE LCh, which serve for color communication and calibration in printing, photography, and display technology.

CIE xyY and the Chromaticity Diagram

The CIE xyY space, also introduced in 1931, separates chromaticity from luminance. Since Y already represents the luminance of the standard observer, it only remains to remove the brightness from the other two dimensions. We normalize the XYZ values by their sum:

$$x = \frac{X}{X+Y+Z}, \qquad y = \frac{Y}{X+Y+Z}, \qquad z = \frac{Z}{X+Y+Z} = 1 - x - y \tag{0.9}$$

The third coordinate z is redundant, so x and y describe the chromaticity and Y the luminance. To reverse the transformation from x, y, and Y, we compute

$$X = \frac{Y}{y}\,x, \qquad Z = \frac{Y}{y}\,(1 - x - y) \tag{0.10}$$

Doubling the light doubles X, Y, and Z but leaves x and y unchanged. This invariance is what makes chromaticity useful whenever brightness should be ignored.

Figure 0.9 shows the chromaticity diagram of the xyY space, with the names of the color regions. The wavelengths along the spectral locus are given in micrometers (0.38 to 0.65), and the color temperatures of the Planckian locus are marked at the bottom. The diagram is rendered in sRGB, so colors outside the sRGB triangle of Figure 0.10 are not displayed accurately.

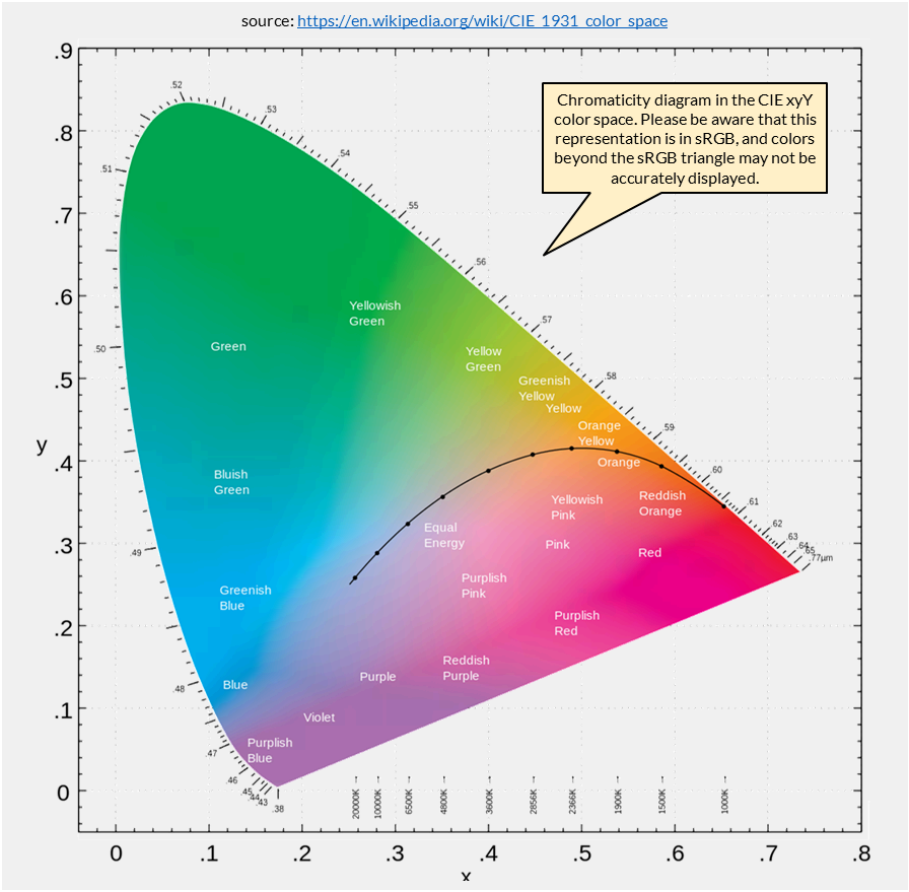


Figure 0.9: The chromaticity diagram of CIE xyY with named color regions and the equal-energy point.

Example

Spectrum A of the previous example has $X = Y = Z = 100$, so $x = y = 100/300 = 1/3$. This is the **equal-energy point** in the middle of Figure 0.9. The D65 white of sRGB, which we meet below, has $(X, Y, Z) = (0.9505, 1.0000, 1.0890)$ and therefore

$$x = \frac{0.9505}{3.0395} = 0.3127, \qquad y = \frac{1.0000}{3.0395} = 0.3290 \tag{0.11}$$

Going back from $(x, y, Y) = (0.3127, 0.3290, 1)$ gives $X = 0.3127/0.3290 = 0.9505$ and $Z = (1 - 0.3127 - 0.3290)/0.3290 = 1.0891$, which matches the original values up to rounding.

The CIE xyY space covers all colors that the standard observer sees, and it is a widely used color specification. It describes color as the standard observer sees it, not the inherent color of an object: the perceived color of an object changes with the lighting and with the color temperature of the light source. In low light, the human eye cannot see color at all and is limited to shades of gray. By including all colors of the standard observer, xyY links human perception to digital color systems and supports consistent color reproduction across devices.

sRGB and Linear RGB

RGB is the main color model in computing. HP and Microsoft proposed sRGB in 1996 as an additive standard color space for monitors, printers, and the Internet. It became the standard IEC 61966-2-1:1999, and software assumes it whenever an image does not specify its color space.

sRGB takes its primaries from the HDTV standard ITU-R BT.709, also called Rec. 709. They define its color gamut, the range of colors it can produce. The primaries became widely used because they matched the CRT monitors of the time. The chromaticities of the primaries and the white point are:

	x	y
Red	0.6400	0.3300
Green	0.3000	0.6000
Blue	0.1500	0.0600
White (D65)	0.3127	0.3290

With non-negative channel values, all sRGB colors lie inside the triangle spanned by the primaries in Figure 0.10. This gamut does not cover all chromaticities; it misses a large part of the greens and of the blue-green range.

source: https://en.wikipedia.org/wiki/SRGB

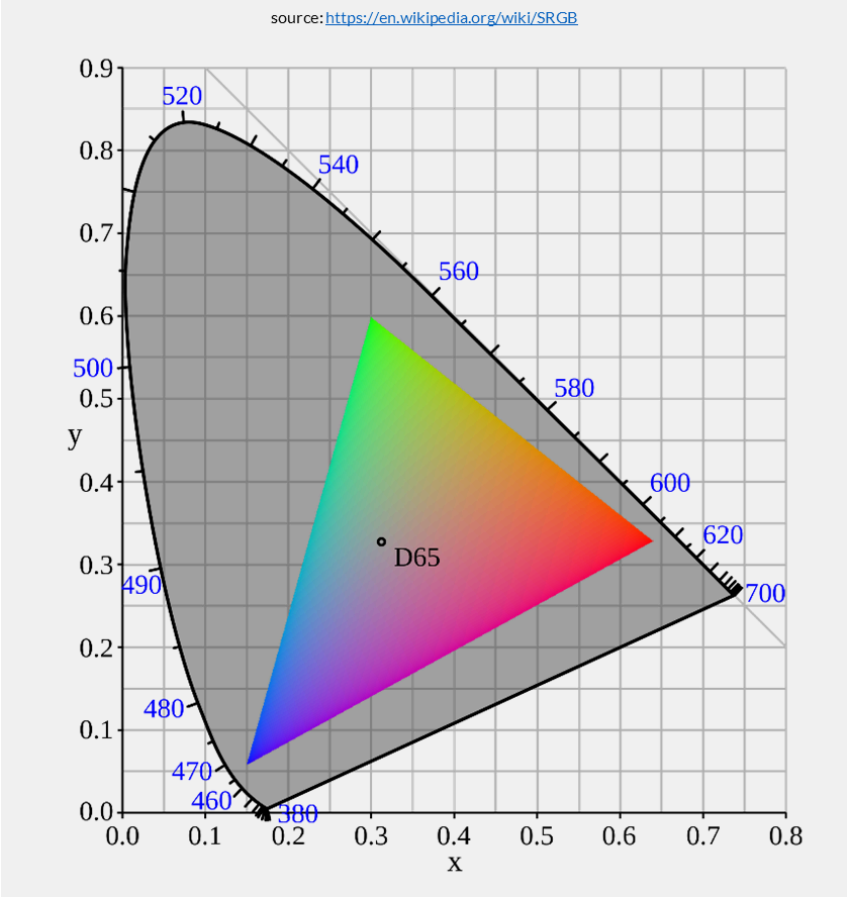


Figure 0.10: The sRGB gamut as a triangle in the chromaticity diagram, with the D65 white point.

The channel values of sRGB are not proportional to the light. They are encoded nonlinearly, with an approximate gamma of 2.2. Let c_{sRGB} denote a channel value (red, green, or blue) in sRGB and c_{linear} the corresponding value in linear RGB, which is proportional to the light. Both range from 0 to 1. For quantized values, we first divide by $2^{\text{bits}} - 1$, and after the conversion we multiply again.

Key Formula: sRGB Transfer Functions

$$c_{\text{linear}} = \tag{0.12}$$

$$c_{\text{sRGB}} = \tag{0.13}$$

A short linear segment near black and a power curve everywhere else. Together they stay close to a pure power law with exponent 2.2. The first function decodes sRGB to linear light, the second encodes linear light as sRGB.

Example

We decode the running color (255, 200, 100). Dividing by 255 gives $(1.0000, 0.7843, 0.3922)$. All three values exceed 0.04045, so the power branch applies:

$$\begin{aligned} r_{\text{linear}} &= \left(\frac{1.0000 + 0.055}{1.055}\right)^{2.4} = 1.0000^{2.4} = 1.0000 \\ g_{\text{linear}} &= \left(\frac{0.7843 + 0.055}{1.055}\right)^{2.4} = 0.7956^{2.4} = 0.5776 \\ b_{\text{linear}} &= \left(\frac{0.3922 + 0.055}{1.055}\right)^{2.4} = 0.4238^{2.4} = 0.1274 \end{aligned} \tag{0.14}$$

The linear values are $(1.0000, 0.5776, 0.1274)$. The blue channel uses 39 percent of the code range but carries only 13 percent of the light. A pure power law with exponent 2.2 would give $0.7843^{2.2} = 0.5860$ for green, close to the exact 0.5776.

The nonlinear encoding is not a defect. The middle code value 128 corresponds to only 21.6 percent of the light, yet it looks like a medium gray, with a CIE Lab lightness of 53.6 out of 100. The encoding spends more code values on dark tones, where the eye distinguishes small differences, and fewer on bright tones. This is why 8 bits per channel suffice for most images.

The constant is 0.055

The encoding subtracts 0.055, not 0.05. With 0.055, white stays white ($1.055 - 0.055 = 1$), and both branches give the same value 0.04045 at the threshold 0.0031308. With 0.05, white would map to 1.005 and the curve would jump at the threshold. The same care applies to the decoding threshold 0.04045, which is $12.92 \cdot 0.0031308$.

The conversion between CIE XYZ and linear RGB is a matrix multiplication. The sRGB standard defines the matrix from linear RGB to XYZ with four digits, and its inverse from XYZ to linear RGB:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.4124 & 0.3576 & 0.1805 \\ 0.2126 & 0.7152 & 0.0722 \\ 0.0193 & 0.1192 & 0.9505 \end{bmatrix} \begin{bmatrix} r_{\text{linear}} \\ g_{\text{linear}} \\ b_{\text{linear}} \end{bmatrix} \tag{0.15}$$

$$\begin{bmatrix} r_{\text{linear}} \\ g_{\text{linear}} \\ b_{\text{linear}} \end{bmatrix} = \begin{bmatrix} 3.2406255 & -1.5372080 & -0.4986286 \\ -0.9689307 & 1.8757561 & 0.0415175 \\ 0.0557101 & -0.2040211 & 1.0569959 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \tag{0.16}$$

Each column of the first matrix holds the XYZ values of one primary at full intensity, and the second row holds their luminance: green contributes 0.7152 of the luminance of white, red 0.2126, and blue only 0.0722. For white, $r = g = b = 1$, the first matrix gives the row sums $(X, Y, Z) = (0.9505, 1.0000, 1.0890)$, the D65 white point scaled to $Y = 1$. Some libraries, such as OpenCV, use six-digit variants of these matrices; for the running color, they change X, Y, and Z by less than 0.0001.

Both matrices work on linear values. To go from sRGB to XYZ, we first decode with the transfer function and then multiply with the first matrix. To go from XYZ to sRGB, we multiply with the second matrix and then encode. Since the RGB space does not cover the whole XYZ space, the second matrix can produce values outside the range from 0 to 1. The simplest remedy is clipping: we set negative values to 0 and values greater than 1 to 1.

For the running color, the first matrix gives

$$\begin{aligned} X &= 0.4124 \cdot 1.0000 + 0.3576 \cdot 0.5776 + 0.1805 \cdot 0.1274 = 0.4124 + 0.2065 + 0.0230 = 0.6419 \\ Y &= 0.2126 \cdot 1.0000 + 0.7152 \cdot 0.5776 + 0.0722 \cdot 0.1274 = 0.2126 + 0.4131 + 0.0092 = 0.6349 \\ Z &= 0.0193 \cdot 1.0000 + 0.1192 \cdot 0.5776 + 0.9505 \cdot 0.1274 = 0.0193 + 0.0688 + 0.1211 = 0.2093 \end{aligned} \tag{0.17}$$

The luminance $Y = 0.6349$ is dominated by the red and green channels. Its chromaticity is $x = 0.6419/1.4861 = 0.4320$ and $y = 0.6349/1.4861 = 0.4272$.

RGB values are usually quantized into integer ranges by a simple multiplication with $2^{\text{bits}} - 1$. True color uses 8 bits per channel, so the multiplier is 255. Formats with 16 bits per channel use 65535, and other deep color formats use 10 or 12 bits per channel. Some formats quantize with a set of reference colors, a **color palette**, instead; GIF images, for example, have at most 256 palette colors. A pixel then stores the index of its nearest palette color.

Wide-Gamut RGB Spaces

Apart from sRGB and linear RGB, many other RGB spaces exist. We can create an RGB space simply by choosing three primaries and a white point. Alternative spaces extend the limited gamut of sRGB to a

wider range of colors. For example, Rec. 2020, the ITU-R recommendation BT.2020 for ultra-high-definition television (UHDTV), uses the D65 white point with the primaries red $(0.708, 0.292)$, green $(0.170, 0.797)$, and blue $(0.131, 0.046)$. They lie on or near the spectral locus, so Rec. 2020 has a much wider gamut than HDTV based on Rec. 709.

Figure 0.11 compares the gamuts of six RGB spaces. sRGB and Rec. 709 share the same primaries and white point and differ only in their transfer functions. DCI-P3 and Adobe RGB extend the gamut mainly toward green. ProPhoto RGB goes beyond the spectral locus: two of its primaries are imaginary colors, chosen to include more of the green and blue regions.

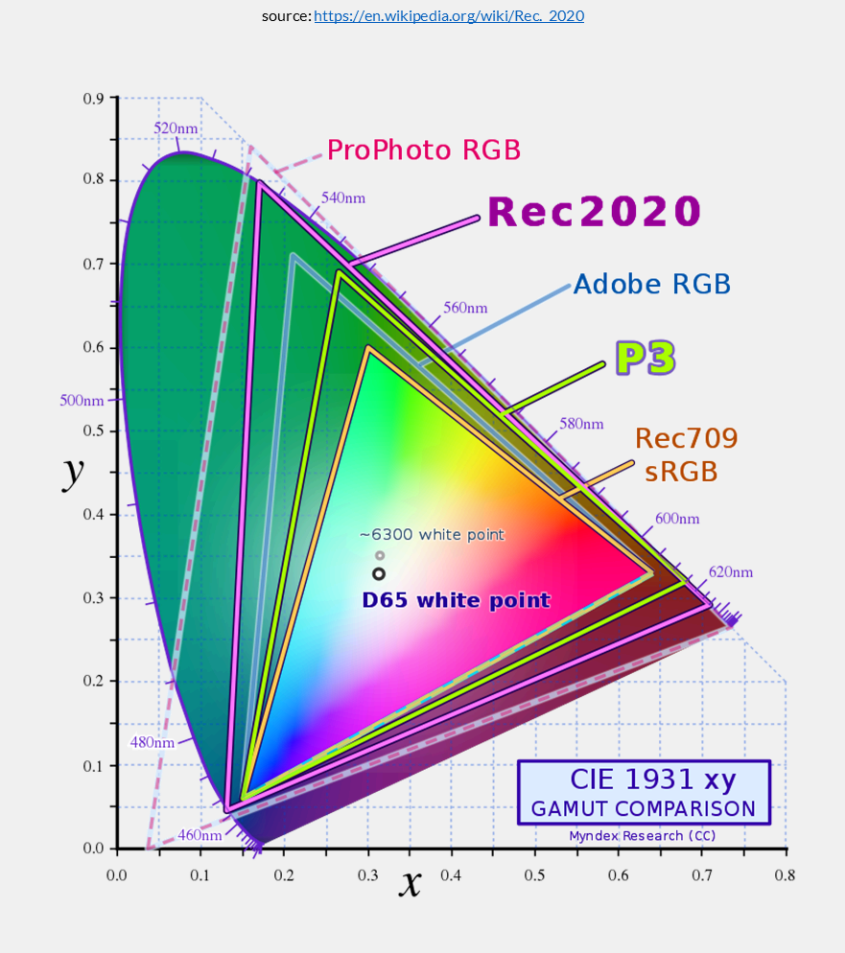


Figure 0.11: Gamuts of sRGB, Rec. 709, P3, Adobe RGB, Rec. 2020, and ProPhoto RGB in the chromaticity diagram.

The same triple of numbers means different colors in different RGB spaces. In sRGB, the value $(0, 255, 0)$ is the most saturated green the space can show. In Rec. 2020, the same value is a much more saturated green. To move an image from one RGB space to another, we decode it to linear values, convert to XYZ with the matrix of the source space, and convert back with the inverse matrix of the target space.

CIE Lab

The CIE xyY space covers the entire range of visible colors, but it is not perceptually uniform: two pairs of colors with the same distance can look very different depending on where they lie in the space. The CIE Lab color space, defined in 1976 and also written CIELAB or L*a*b*, was created to provide a more perceptually uniform scale. It covers all visible colors, so its gamut exceeds that of RGB spaces, and it is device independent. It is therefore commonly used to map colors between different systems.

- The component L^* represents **lightness**. It is derived from the luminance Y but adjusted for perception, so that a change of one unit corresponds to roughly the same perceived change in lightness everywhere on the scale. L^* ranges from 0 for black to 100 for white.

- The component a^* represents the **red-green opponent axis**. Negative values indicate green, positive values indicate red or magenta.
- The component b^* represents the **blue-yellow opponent axis**. Negative values indicate blue, positive values indicate yellow.

When both a^* and b^* are 0, the color is a neutral gray. The two axes are unbounded in principle. For sRGB colors, a^* ranges from about −86 for the green primary to about 98 for magenta, and b^* from about −108 for the blue primary to about 94 for yellow. Implementations that store Lab in 8 bits clamp a^* and b^* to the range from −128 to 127.

Figure 0.12 shows three slices through the sRGB gamut in Lab at the lightness values 75, 50, and 25. The range of chromatic colors is widest at medium lightness and shrinks toward white and black.

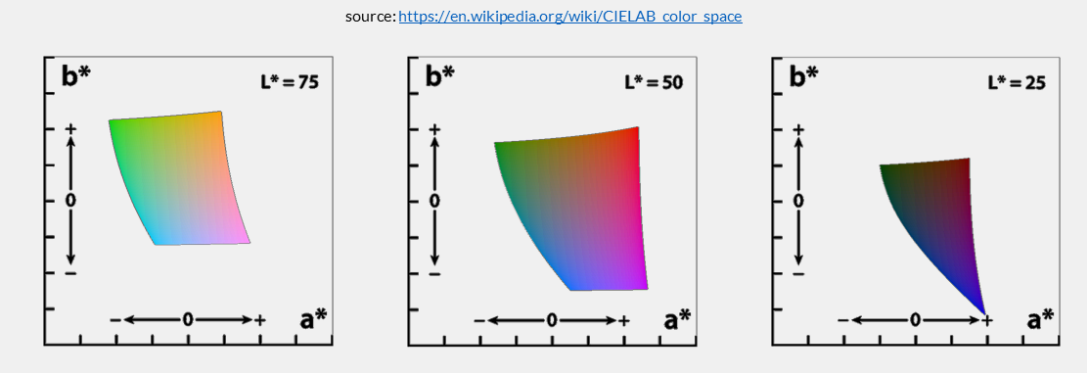


Figure 0.12: Slices through the sRGB gamut in CIE Lab at $L^* = 75$, 50, and 25.

Key Formula: CIE Lab

$$L^* = 116 \cdot \left(f\frac{Y}{Y_n}\right) - 16$$

$$a^* = 500 \cdot \left(\left(f\frac{X}{X_n}\right) - \left(f\frac{Y}{Y_n}\right)\right) \tag{0.18}$$

$$b^* = 200 \cdot \left(\left(f\frac{Y}{Y_n}\right) - \left(f\frac{Z}{Z_n}\right)\right)$$

$$f(t) = \tag{0.19}$$

Lab compares each tristimulus value with the white point (X_n, Y_n, Z_n) and compresses the ratio with a cube root, which mimics the nonlinear response of perception. Lightness comes from Y alone; the differences between the compressed values give the red-green and the blue-yellow axes.

The constants in $f(t)$ have simple meanings. The threshold is $(6/29)^3 = 0.008856$, and below it a straight line with slope $841/108 = 7.787$ and offset $4/29 = 0.1379$ replaces the cube root. The line meets the cube root at the threshold with the same slope, and it avoids the infinite slope of the cube root at 0. For sRGB under D65 with Y scaled to 1, the white point is $(X_n, Y_n, Z_n) = (0.9505, 1.0000, 1.0890)$. Only the ratios X/X_n, Y/Y_n, and Z/Z_n enter the formula, so any scale works as long as the white point uses the same scale; on a scale from 0 to 255, the white point becomes $(242.38, 255.0, 277.70)$. Published values for D65 differ in the fourth digit, such as $(0.95047, 1.00000, 1.08883)$; for the running color, this changes a^* and b^* by less than 0.01.

Example

The running color has $(X, Y, Z) = (0.6419, 0.6349, 0.2093)$. The ratios to the white point are

$$\frac{X}{X_n} = \frac{0.6419}{0.9505} = 0.6754, \qquad \frac{Y}{Y_n} = \frac{0.6349}{1.0000} = 0.6349, \qquad \frac{Z}{Z_n} = \frac{0.2093}{1.0890} = 0.1922 \tag{0.20}$$

All three exceed 0.008856, so f is the cube root: $f(0.6754) = 0.8774$, $f(0.6349) = 0.8595$, and $f(0.1922) = 0.5771$. Then

$$\begin{aligned} L^* &= 116 \cdot 0.8595 - 16 = 83.70 \\ a^* &= 500 \cdot (0.8774 - 0.8595) = 8.95 \\ b^* &= 200 \cdot (0.8595 - 0.5771) = 56.48 \end{aligned} \tag{0.21}$$

The color is light ($L^* = 83.70$), slightly red ($a^* = 8.95$), and strongly yellow ($b^* = 56.48$): a warm light orange, as expected.

Lab measures the difference between two colors as the Euclidean distance of their coordinates, $\Delta E^*_{ab} = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2}$, a formula from 1976 also called CIE76. The four boxes of \#image-features-color, (155, 200, 100), (255, 200, 100), (255, 100, 100), and (255, 0, 100), differ by 100 units in one sRGB channel between neighbors. In Lab, the three steps measure 42.7, 59.8, and 27.7. Equal steps in sRGB are thus far from equal steps in perception, and Lab makes the difference visible in the numbers.

Lab is only approximately uniform. It is known to be least uniform for blue hues, and later formulas such as CIEDE2000 correct the Euclidean distance with weights for lightness, chroma, and hue. For color histograms and similar features, the plain Euclidean distance in Lab is usually good enough and much cheaper.

HSL and HSV

Artists typically begin with a relatively bright color and then mix it:

- with white to **tint** the color,
- with black to **shade** the color, or
- with both white and black, that is gray, to **tone** the color.

To support these techniques in computer graphics, the HSL and HSV color models provide alternative representations of RGB that make color manipulation easier. Both were described in 1978 in the same issue of the journal *Computer Graphics*: HSV by Alvy Ray Smith, and HSL by Joblove and Greenberg. Both models use hue and chroma to define the chromaticity. Figure 0.13 shows their geometry. HSL adds **lightness** and forms a double cone. It places the fully saturated colors at $L = 1/2$ and allows tinting toward $L = 1$ and shading toward $L = 0$ without changing the saturation. HSV adds **value** and forms a single cone. It places the fully saturated colors at $V = 1$ and allows shading toward $V = 0$ without changing the saturation, while tinting does change the saturation. The hexagon on the right shows the hue plane: the six primary and secondary colors sit at its corners, 60 degrees apart.

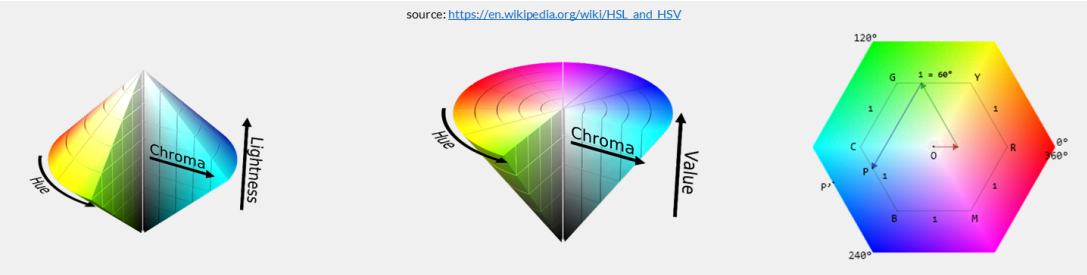


Figure 0.13: HSL as a double cone (left), HSV as a cone (center), and the hexagonal hue plane (right).

Both models start from the largest and the smallest channel value and their difference, the chroma:

$$M = \max(R, G, B), \qquad m = \min(R, G, B), \qquad C = M - m \tag{0.22}$$

The hue depends on which channel is the largest. It is undefined for grays with $C = 0$ and set to 0 by convention:

$$H' = \qquad H = 60^\circ \cdot H' \tag{0.23}$$

Value and lightness, and the saturation of each model, are

$$V = M, \qquad S_{\text{HSV}} = \tag{0.24}$$

$$L = \frac{1}{2}(M + m), \qquad S_{\text{HSL}} = \tag{0.25}$$

In practice, libraries compute HSV and HSL directly from the gamma-encoded sRGB values with R, G, and B between 0 and 1.

Example

For the running color, $(R, G, B) = (1.0000, 0.7843, 0.3922)$. Then $M = 1.0000$, $m = 0.3922$, and $C = 0.6078$. Red is the largest channel, so

$$H' = \left(\frac{0.7843 - 0.3922}{0.6078}\right) \bmod 6 = 0.6452, \qquad H = 60^\circ \cdot 0.6452 = 38.7^\circ \tag{0.26}$$

The hue lies between red at 0° and yellow at 60°, closer to yellow. For HSV, $V = 1.0000$ and $S_{\text{HSV}} = 0.6078/1.0000 = 0.608$. For HSL, $L = (1.0000 + 0.3922)/2 = 0.6961$ and

$$S_{\text{HSL}} = \frac{0.6078}{1 - |\,2 \cdot 0.6961 - 1\,|} = \frac{0.6078}{0.6078} = 1.000 \tag{0.27}$$

The same color has a saturation of 0.61 in HSV and of 1.00 in HSL. The two models define saturation differently, so their saturation values cannot be compared.

HSL and HSV separate hue from brightness, which makes them convenient for color pickers and for features that should ignore brightness. They are not perceptually uniform, however. Pure yellow (255, 255, 0) and pure blue (0, 0, 255) both have $V = 1$ and $S_{\text{HSV}} = 1$, yet their Lab lightness is 97.1 and 32.3. Value and lightness in these models follow the symmetry of the RGB cube, not human perception.

One Color Through All Spaces

The following table collects the coordinates of the running color, together with the same surface in a shadow that receives half the light. The shadow halves the linear RGB values; rounded to 8 bits, its sRGB value is (188, 146, 71). The line from the white point through the chromaticity of the color hits the spectral locus at 579 nm, a yellow, and the color lies at 0.61 of the distance from the white point to the locus: its dominant wavelength is 579 nm and its saturation in the sense of the chromaticity diagram is 0.61.

Space	Full light	Half light (shadow)
sRGB	(255, 200, 100)	(187.5, 146.3, 71.4)
Linear RGB	(1.0000, 0.5776, 0.1274)	(0.5000, 0.2888, 0.0637)
CIE XYZ	(0.6419, 0.6349, 0.2093)	(0.3210, 0.3174, 0.1046)
Chromaticity (x, y)	(0.4320, 0.4272)	(0.4320, 0.4272)
CIE Lab (L^*, a^*, b^*)	(83.70, 8.95, 56.48)	(63.13, 7.10, 44.83)
Lab hue angle and chroma	81.0°, 57.2	81.0°, 45.4
HSV (H, S, V)	(38.7°, 0.608, 1.000)	(38.7°, 0.619, 0.735)

The chromaticity (x, y) does not change at all, because halving the light halves X, Y, and Z alike. The hue of HSV and the hue angle in the a^*b^* plane do not change either. This is the property that Chapter 12 uses to remove shadows in motion detection: a shadow darkens the background but keeps its chromaticity, while a moving object usually changes it. Real shadows are less ideal, because they are lit by the bluish light of the sky instead of the sun, so their chromaticity shifts slightly, and a threshold has to tolerate this shift.

Lab chroma is not brightness invariant

Lab separates lightness from chromaticity only approximately. In the table, a^* and b^* shrink in the shadow, and the Lab chroma drops from 57.2 to 45.4, because the cube root compresses all three ratios

> alike. A distance in the a^*b^* plane therefore reports a difference of about 12 units between a surface and its shadow. Features that must ignore brightness should use the chromaticity (x, y), the hue angle, or normalized RGB values instead.

Extended Image Features

Where this is used

\#image-features-color compares images with color histograms, the quadratic distance, and color moments; this section adds the one-pass computation of moments and the transformation that turns the quadratic distance into a Euclidean distance. \#image-features-edges-texture-shape uses Sobel gradients, gradient histograms, Gabor filters, and Gabor moments; this section adds gradient moments, Laws' texture energy, and the computation of Gabor filters in Fourier space. Distance and Similarity Measures treats the quadratic distance as a weighted distance and derives its eigenvalue decomposition.

The image features of \#image-features are defined by formulas that are easy to read but not always cheap to compute. A color moment needs the mean before it can compute the variance. A quadratic histogram distance multiplies a difference vector with an $N \times N$ matrix for every comparison. A Gabor filter bank convolves the image with a dozen large kernels. This section collects what the main text leaves out: how to compute moments in a single pass, how to replace the quadratic distance with a Euclidean one, two further texture features, namely gradient moments and Laws' texture energy, and how Gabor filters work in Fourier space. Each subsection starts with a short reminder of the definition it builds on.

Moments in One Pass

For a channel c of an image with N rows and M columns, \#image-features-color defines the mean μ_c, the standard deviation σ_c, the skewness s_c, and the kurtosis k_c:

$$\mu_c = \frac{1}{NM} \sum_{x,y} c(x,y), \qquad \sigma_c^2 = \frac{1}{NM} \sum_{x,y} \big(c(x,y) - \mu_c \big)^2 \tag{0.28}$$

$$s_c = \frac{1}{NM} \sum_{x,y} \left(\frac{c(x,y) - \mu_c}{\sigma_c} \right)^3, \qquad k_c = \frac{1}{NM} \sum_{x,y} \left(\frac{c(x,y) - \mu_c}{\sigma_c} \right)^4 \tag{0.29}$$

The covariance of two channels c_i and c_j follows the same pattern:

$$\mathrm{cov}_{c_i,c_j} = \frac{1}{NM} \sum_{x,y} \big(c_i(x,y) - \mu_{c_i} \big) \big(c_j(x,y) - \mu_{c_j} \big) \tag{0.30}$$

Written like this, every formula except the mean needs μ_c before it can start. A direct implementation reads all pixels twice: once for the mean, once for the other moments. For a large collection, the second pass doubles the reading time, and for a stream of video frames it means buffering every frame.

The way out is to accumulate raw moments instead of central ones. For $1 \le n \le 4$, the raw moment $a_{c,n}$ averages the n-th power of the channel values, and $b_{i,j}$ averages the product of two channels:

$$a_{c,n} = \frac{1}{NM} \sum_{x,y} c(x,y)^n, \qquad b_{i,j} = \frac{1}{NM} \sum_{x,y} c_i(x,y)\, c_j(x,y) \tag{0.31}$$

All of these sums can be updated pixel by pixel in one pass. The central moments follow from the binomial expansion. For the third power, $(c-\mu)^3 = c^3 - 3c^2\mu + 3c\mu^2 - \mu^3$. Averaging over all pixels and substituting $\mu = a_{c,1}$ gives $a_{c,3} - 3a_{c,1}a_{c,2} + 3a_{c,1}^3 - a_{c,1}^3 = a_{c,3} - 3a_{c,1}a_{c,2} + 2a_{c,1}^3$. The fourth power works the same way.

Key Formula: One-Pass Moments

$$\begin{aligned} \mu_c &= a_{c,1} \\ \sigma_c^2 &= a_{c,2} - a_{c,1}^2 \\ s_c &= \frac{a_{c,3} - 3a_{c,1}a_{c,2} + 2a_{c,1}^3}{\sigma_c^3} \\ k_c &= \frac{a_{c,4} - 4a_{c,1}a_{c,3} + 6a_{c,1}^2a_{c,2} - 3a_{c,1}^4}{\sigma_c^4} \\ \operatorname{cov}_{c_i,c_j} &= b_{i,j} - \mu_{c_i}\,\mu_{c_j} \end{aligned} \tag{0.32}$$

Four running sums per channel and one per pair of channels are enough. All moments follow at the end from a few multiplications.

Example

Take eight pixels with lightness values $L = 20, 40, 40, 40, 50, 50, 70, 90$ and a values $5, 10, 0, 10, 15, 5, 20, 15$. One pass over the pixels yields the raw moments of L:

$$a_{L,1} = 50, \quad a_{L,2} = 2900, \quad a_{L,3} = 190,250, \quad a_{L,4} = 13,745,000 \tag{0.33}$$

The mean is $\mu_L = 50$ and the variance is $\sigma_L^2 = 2900 - 50^2 = 400$, so the standard deviation is $\sigma_L = 20$. The third central moment is $190,250 - 3 \cdot 50 \cdot 2900 + 2 \cdot 50^3 = 5250$, and the skewness is $s_L = 5250/20^3 = 5250/8000 = 0.65625$. The fourth central moment is

$$13,745,000 - 38,050,000 + 43,500,000 - 18,750,000 = 445,000 \tag{0.34}$$

and the kurtosis is $k_L = 445,000/20^4 = 445,000/160,000 = 2.78125$. The two-pass definitions give exactly the same values. The positive skewness reflects the long tail toward the bright pixels 70 and 90; a kurtosis below 3 means the distribution is flatter than a normal distribution.

For the covariance, the mean of a is 10 and the average product is $b_{L,a} = 581.25$, so $\operatorname{cov}_{L,a} = 581.25 - 50 \cdot 10 = 81.25$.

The kurtosis defined here equals 3 for a normal distribution. Many libraries report the excess kurtosis $k_c - 3$ instead; for the eight pixels above, `scipy.stats.kurtosis` returns –0.21875 by default. Both conventions work as features, as long as all images use the same one.

One-pass formulas cancel large numbers

In the example, the four terms of the fourth moment are tens of millions, but the result is 445,000. Most of the value cancels, and rounding errors in the accumulated sums survive the subtraction. On the Matterhorn photo of \#image-features-color (1,091,840 pixels, mean lightness 60.9), a simple loop that accumulates the four sums in 32-bit floating point gives a variance that is 0.39% too low and a kurtosis that is 3.2% too high. Three remedies help. Accumulate in 64-bit floating point: the relative errors then stay below 10^{-10}. Subtract a rough estimate of the mean before accumulating, since the standard deviation, skewness, kurtosis, and covariance do not change when all values shift: subtracting 85, the mean of the first ten rows, reduces the 32-bit errors below 0.05%. Or use update formulas that keep running central moments instead of raw sums; they are numerically stable and can also merge partial results computed in parallel (see Pébay in the Further Reading).

Fast Quadratic Distances

Let $\boldsymbol{h}$ and $\boldsymbol{g}$ be the normalized histograms of two images over N reference colors, so that the entries of each sum to 1. \#image-features-color compares them with the quadratic distance, which takes the similarity between reference colors into account through a matrix $\boldsymbol{A}$. Its entries derive from the distances $d_{i,j}$ between the reference colors c_i and c_j:

$$\delta_{\text{quad}}(\boldsymbol{h},\boldsymbol{g}) = (\boldsymbol{h}-\boldsymbol{g})^\top \boldsymbol{A}\,(\boldsymbol{h}-\boldsymbol{g}), \qquad a_{i,j} = 1 - \frac{d_{i,j}}{\max_{k,l} d_{k,l}} \tag{0.35}$$

Each comparison costs N^2 multiplications, against N for a Euclidean distance: 4096 for 64 reference colors, and a million for 1000. The index structures of Vector Search also expect a Euclidean or similar standard distance, not an arbitrary matrix. Both problems disappear if we can transform the histograms so that a Euclidean distance yields the same values.

The transformation uses the eigenvalue decomposition of $\boldsymbol{A}$. A symmetric matrix can be written as $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$, where the columns $\boldsymbol{u}_i$ of $\boldsymbol{U}$ are orthonormal eigenvectors and $\boldsymbol{\Lambda}$ is the diagonal matrix of the eigenvalues λ_i. If all eigenvalues are nonnegative, the matrix is **positive semidefinite**, and the square root $\boldsymbol{\Lambda}^{1/2}$ exists. Splitting $\boldsymbol{\Lambda}$ into two square roots gives

$$(\boldsymbol{h}-\boldsymbol{g})^\top \boldsymbol{U}\boldsymbol{\Lambda}^{1/2}\,\boldsymbol{\Lambda}^{1/2}\boldsymbol{U}^\top(\boldsymbol{h}-\boldsymbol{g}) = \parallel \boldsymbol{\Lambda}^{1/2}\boldsymbol{U}^\top(\boldsymbol{h}-\boldsymbol{g}) \parallel^2 \tag{0.36}$$

Distance and Similarity Measures shows the geometry of this rotation and scaling.

Key Formula: Quadratic Distance as a Euclidean Distance

$$\boldsymbol{h}' = \boldsymbol{\Lambda}^{1/2}\boldsymbol{U}^\top\boldsymbol{h}, \qquad \delta_{\text{quad}}(\boldsymbol{h},\boldsymbol{g}) = \parallel \boldsymbol{h}' - \boldsymbol{g}' \parallel^2 = \sum_{i=1}^{N} \lambda_i \left(\boldsymbol{u}_i^\top(\boldsymbol{h}-\boldsymbol{g})\right)^2 \tag{0.37}$$

Condition: $\boldsymbol{A}$ is symmetric and positive semidefinite, that is, $\lambda_i \geq 0$ for all i. The transformation rotates the histogram into the axes of the eigenvectors and stretches each axis by $\sqrt{\lambda_i}$. Every histogram is transformed once when it is indexed; at query time, a plain squared Euclidean distance replaces the matrix product.

Example

Recall the three reference colors sky blue, pale blue, and dark green of the hazy-sky example in \#image-features-color. Their matrix entries are $a_{1,2} = 0.713$, $a_{1,3} = 0.099$, and $a_{2,3} = 0$:

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0.713 & 0.099 \\ 0.713 & 1 & 0 \\ 0.099 & 0 & 1 \end{pmatrix} \tag{0.38}$$

The eigenvalues are 0.2802, 1, and 1.7198, all positive. The eigenvector of the smallest eigenvalue is $\boldsymbol{u}_1 = (0.7071, -0.7004, -0.0972)$: it contrasts sky blue with pale blue. The second, $\boldsymbol{u}_2 = (0, -0.1375, 0.9905)$, is almost pure dark green, and the third, $\boldsymbol{u}_3 = (0.7071, 0.7004, 0.0972)$, combines the two blues.

The query photo $\boldsymbol{h} = (0.6, 0, 0.4)$, the hazy photo $\boldsymbol{g} = (0, 0.6, 0.4)$, and the forest photo $\boldsymbol{k} = (0.2, 0, 0.8)$ map to

$$\boldsymbol{h}' = (0.2040, 0.3962, 0.6074), \quad \boldsymbol{g}' = (-0.2430, 0.3137, 0.6021), \quad \boldsymbol{k}' = (0.0337, 0.7924, 0.2873) \tag{0.39}$$

The squared Euclidean distances between the transformed vectors are $\parallel \boldsymbol{h}' - \boldsymbol{g}' \parallel^2 = 0.207$ and $\parallel \boldsymbol{h}' - \boldsymbol{k}' \parallel^2 = 0.288$, the same values as the quadratic distances of the main text. Of the first distance, 0.200 lies on the first axis, the one with the smallest eigenvalue: moving pixels between sky blue and pale blue is cheap.

The transformed coordinates also give a fast filter step. Because every term $\lambda_i(\boldsymbol{u}_i^\top(\boldsymbol{h}-\boldsymbol{g}))^2$ is nonnegative, dropping some axes can only reduce the sum, so a distance over the axes with the largest eigenvalues is a lower bound of the full distance. A search can compute this short distance for all candidates, discard those whose lower bound already exceeds the current best, and compute the full distance only for the rest, without ever discarding a true neighbor. The bound can be loose, though. Keeping the two largest eigenvalues in the example gives 0.259 for the forest photo (true value 0.288) but only 0.007 for the hazy photo (true value 0.207), because $\boldsymbol{h} - \boldsymbol{g}$ lies almost entirely on the dropped axis.

When the Matrix Is Not Positive Semidefinite

The construction $a_{i,j} = 1 - d_{i,j} / \max d$ makes $\boldsymbol{A}$ symmetric, but it does not guarantee nonnegative eigenvalues. With three reference colors and any distance that satisfies the triangle inequality, $\boldsymbol{A}$ is always positive semidefinite, as in the example above; the problem appears with larger palettes. Take seven reference colors in RGB with channels scaled to $[0, 1]$: a mid gray $(0.5, 0.5, 0.5)$ and the six colors that differ from it by 0.5 in exactly one channel, up or down. Under the Manhattan distance, a common way to combine the hue angle difference with the other channel differences in HSV, each outer color lies at distance 0.5 from the gray and at distance 1 from every other outer color. The matrix has 1 on the diagonal, 0.5 between the gray and each outer color, and 0 between outer colors. Its smallest eigenvalue is $1 - 0.5\sqrt{6} = -0.2247$, so $\sqrt{\lambda}$ does not exist for this axis.

The negative eigenvalue has a visible effect on unnormalized counts. For $\boldsymbol{h} = (3, 0, 0, 0, 0, 0, 0)$, three gray pixels, and $\boldsymbol{g} = (0, 1, 1, 1, 1, 1, 1)$, one pixel of each outer color, the "distance" is $\delta_{\text{quad}}(\boldsymbol{h}, \boldsymbol{g}) = -3$. After normalization, the same two histograms have the distance 0.1667. The difference of two normalized histograms always sums to zero, and on such vectors this matrix stays nonnegative. This holds in general for Euclidean and Manhattan ground distances: for these distances, $\boldsymbol{x}^\top \boldsymbol{A}\, \boldsymbol{x} \geq 0$ for every $\boldsymbol{x}$ whose entries sum to zero. The distance values are therefore fine, but the transformation still needs a square root of every eigenvalue. Three remedies exist:

1. **Center the matrix.** Replace $\boldsymbol{A}$ by $\boldsymbol{A}_c = \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$ with $\boldsymbol{P} = \boldsymbol{I} - \frac{1}{N}\boldsymbol{1}\boldsymbol{1}^\top$. For differences of normalized histograms, $\boldsymbol{A}_c$ yields exactly the same distances as $\boldsymbol{A}$, and for Euclidean and Manhattan ground distances it is positive semidefinite. For the seven colors, its eigenvalues are 0, 0.1429, and five times 1. The eigenvalue 0 belongs to the constant direction $\boldsymbol{1}$, which differences of normalized histograms never use, so the transformed vectors need only $N - 1$ coordinates.
2. **Use a kernel that is positive semidefinite by construction.** With Euclidean distances, the Gaussian weights $a_{i,j} = \exp\!\left(-d_{i,j}^2 / (2s^2)\right)$ with a width s give a positive definite matrix for every set of distinct reference colors. For the seven colors with $s = 0.5$, the smallest eigenvalue is 0.1144. The weights then decrease smoothly with the color distance instead of linearly.
3. **Clip negative eigenvalues to zero.** This is the simplest fix, but it changes all distances, not only the problematic ones. For the seven colors, the distance between the two normalized histograms grows from 0.1667 to 0.3895.

> **Check the eigenvalues before transforming**
>
> A numerical library returns the eigenvalues of any symmetric matrix without complaint. A negative eigenvalue goes unnoticed until its square root turns into "not a number" or a distance becomes negative. Compute the smallest eigenvalue once when $\boldsymbol{A}$ is built, and choose one of the remedies above if it is negative.

Gradient Moments

The Sobel operator of \#image-features-edges-texture-shape assigns two values to every pixel: the gradient magnitude $g_{\text{mag}}(x, y)$ and the gradient direction $g_{\text{dir}}(x, y)$. The main text turns them into an edge density and a gradient histogram. Moments summarize the same two distributions more compactly. For $X \in \{\text{mag}, \text{dir}\}$:

$$\mu_X = \frac{1}{NM} \sum_{x,y} g_X(x, y), \qquad \sigma_X^2 = \frac{1}{NM} \sum_{x,y} \big(g_X(x, y) - \mu_X \big)^2 \tag{0.40}$$

$$s_X = \frac{1}{NM} \sum_{x,y} \left(\frac{g_X(x, y) - \mu_X}{\sigma_X} \right)^3, \qquad k_X = \frac{1}{NM} \sum_{x,y} \left(\frac{g_X(x, y) - \mu_X}{\sigma_X} \right)^4 \tag{0.41}$$

$$\text{cov}_{\text{mag, dir}} = \frac{1}{NM} \sum_{x,y} \big(g_{\text{mag}}(x, y) - \mu_{\text{mag}} \big) \big(g_{\text{dir}}(x, y) - \mu_{\text{dir}} \big) \tag{0.42}$$

Four moments for each quantity and one covariance give nine values that describe the gradient distribution. The mean magnitude measures how much edge energy the image contains. Its skewness is large when most

of the image is flat and a few strong edges stand out. The moments of the direction show whether edges prefer certain directions. All nine values can be computed in one pass with the formulas of the previous subsection.

The gradient direction is an angle

Two gradients at $179°$ and $-179°$ point almost the same way, yet their linear mean is $0°$, the opposite direction. Their circular mean, $\operatorname{atan2}(\overline{\sin\alpha}, \overline{\cos\alpha})$ with the bars denoting averages, is $180°$ as expected. Moments of the raw angle therefore depend on where the angle range starts. Robust alternatives compute the moments of $\cos g_{\text{dir}}$ and $\sin g_{\text{dir}}$, weight each direction by its magnitude, or, when dark-to-light and light-to-dark edges should count the same, double the angle before averaging. The covariance between magnitude and direction inherits the same problem.

Laws' Texture Energy

Kenneth Laws described texture with a small set of 5×5 masks that can be applied to every pixel. The masks derive from four one-dimensional prototype vectors:

$$\boldsymbol{v}_{L5} = (1, 4, 6, 4, 1), \quad \boldsymbol{v}_{E5} = (-1, -2, 0, 2, 1), \quad \boldsymbol{v}_{S5} = (-1, 0, 2, 0, -1), \quad \boldsymbol{v}_{R5} = (1, -4, 6, -4, 1) \tag{0.43}$$

- **L5 (level)** computes a center-weighted local average, close to a Gaussian.
- **E5 (edge)** is a smoothed derivative and responds to step edges.
- **S5 (spot)** has the shape of a Laplacian of Gaussian and responds to small spots.
- **R5 (ripple)** alternates in sign like a small Gabor pattern and responds to fine ripples, the highest frequencies the mask can represent.

The entries of E5, S5, and R5 sum to zero, so these vectors do not respond to regions of constant brightness.

Multiplying a column vector with a row vector gives a two-dimensional kernel. With the first vector running down the rows and the second across the columns, the kernel for E5L5 is

$$\boldsymbol{G}_{E5L5} = \boldsymbol{v}_{E5}^{\top} \boldsymbol{v}_{L5} = \begin{pmatrix} -1 \\ -2 \\ 0 \\ 2 \\ 1 \end{pmatrix} (1 \;\; 4 \;\; 6 \;\; 4 \;\; 1) = \begin{pmatrix} -1 & -4 & -6 & -4 & -1 \\ -2 & -8 & -12 & -8 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 2 & 8 & 12 & 8 & 2 \\ 1 & 4 & 6 & 4 & 1 \end{pmatrix} \tag{0.44}$$

It smooths along each row and differentiates from top to bottom, so it responds to horizontal edges. Its transpose $\boldsymbol{G}_{L5E5}$ responds to vertical edges. The four vectors yield $4 \times 4 = 16$ kernels.

Laws' method convolves the image with each kernel, takes the absolute value of the response, and sums it over a window around each pixel, for example 15 by 15 pixels. The result is one texture energy map per kernel:

$$e_{XY}(x, y) = \sum_{(x', y') \in W(x, y)} \left| \, (I * \boldsymbol{G}_{XY})(x', y') \, \right| \tag{0.45}$$

Three steps reduce the 16 maps to 9:

1. **Drop L5L5.** Its entries sum to 256, so it measures local brightness rather than texture. It can still serve to normalize the other energies for contrast.
2. **Combine the six symmetric pairs.** A pair such as E5L5 and L5E5 measures the same structure in two orientations. Averaging the two energy maps, $e_{E5L5/L5E5} = (e_{E5L5} + e_{L5E5})/2$, gives a measure that treats horizontal and vertical structures alike. The pairs are E5L5/L5E5, S5L5/L5S5, R5L5/L5R5, E5S5/S5E5, E5R5/R5E5, and S5R5/R5S5.
3. **Keep the three symmetric kernels** E5E5, S5S5, and R5R5 as they are.

Of the 16 maps, 15 remain after the first step, and combining six pairs leaves $15 - 6 = 9$ texture energy values $e_i(x, y)$, $1 \le i \le 9$, per pixel.

Average the energy maps, not the kernels

It is tempting to average the two kernels of a pair, $(\boldsymbol{G}_{E5L5} + \boldsymbol{G}_{L5E5})/2$, and convolve only once. The result is a different filter. On a step edge of height 100 that runs from the top right to the bottom left, E5L5 and L5E5 both respond with 3500, and so does the averaged kernel. On an edge from the top left to the bottom right, E5L5 responds with −3500 and L5E5 with +3500: the averaged kernel returns 0 and misses the edge, while the average of the absolute responses is still 3500. Averaging kernels builds a detector for one diagonal; averaging energies treats both diagonals, and horizontal and vertical edges, symmetrically.

From the nine energy maps, we build features with the methods of the main text:

- **Histograms** are possible but large. With four quantization levels per energy, the histogram has $4^9 = 262,144$ bins, far too many for a meaningful feature. With two levels, it has $2^9 = 512$ bins, which is manageable, but each energy is then only "low" or "high", a coarse quantization.
- **Moments** are more compact. Four moments for each of the nine maps give $9 \times 4 = 36$ values, and the covariances of all $\texttt{binom}\ 92 = 36$ pairs of maps add 36 more, for a feature vector of 72 dimensions.

Gabor Filters in Fourier Space

\#image-features-edges-texture-shape introduces the Gabor filter as a Gaussian window multiplied by a complex wave. We repeat its definition because the Fourier view below builds on it. With the coordinates rotated by the angle θ,

$$g_{\lambda,\theta,\phi,\sigma,\gamma}(x,y) = \exp\left(-\frac{\bar{x}^2 + \gamma^2\bar{y}^2}{2\sigma^2}\right)\exp\left(i\left(2\pi\frac{\bar{x}}{\lambda} + \phi\right)\right), \qquad \begin{aligned}\bar{x} &= x\cos\theta + y\sin\theta \\ \bar{y} &= -x\sin\theta + y\cos\theta\end{aligned} \tag{0.46}$$

The wavelength λ sets the frequency $1/\lambda$ of the sinusoid in cycles per pixel, θ its direction, and ϕ its phase. The standard deviation σ sets the size of the Gaussian envelope and γ its aspect ratio. A filter bank combines several directions and scales, and the moments of the filtered images form the texture feature. This subsection shows how the filtering is computed in Fourier space and what the filters look like there.

The FFT Pipeline

The convolution theorem states that a convolution in image space becomes a pointwise multiplication in Fourier space: $I * g = \mathcal{F}^{-1}\left(\ \mathcal{F}(I) \cdot \mathcal{F}(g)\ \right)$. Figure 0.14 shows the resulting pipeline. The fast Fourier transform (FFT) turns the grayscale image into a complex array with a real and an imaginary component. We multiply this array element by element with the Fourier transform of the Gabor filter, and the inverse FFT returns the filtered image. The filtered image is complex as well; its magnitude measures how strongly the selected frequency and direction are present at each pixel. For a bank of filters, the forward FFT of the image is computed only once and shared by all filters.

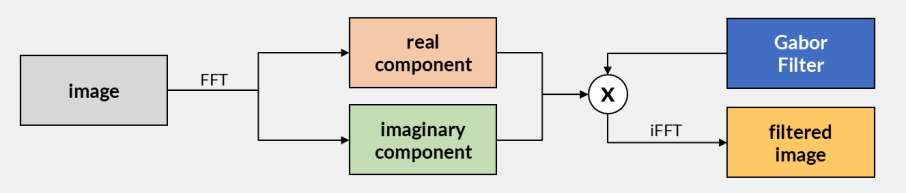


Figure 0.14: Gabor filtering in Fourier space: FFT of the image, pointwise multiplication with the filter, inverse FFT.

To inspect a spectrum, we display $\log\left(\ 1 + |\ \mathcal{F}(I)\ |^2\ \right)$, the logarithm of the energy per frequency, with the zero frequency shifted to the center. The logarithm compresses the large differences between the dominant low frequencies and the weak high ones.

What a Gabor Filter Looks Like in Fourier Space

The Fourier transform of a Gaussian is again a Gaussian, with inverse width. The complex sinusoid $e^{i2\pi\bar{x}/\lambda}$ shifts this Gaussian away from the origin to the frequency $1/\lambda$ in the direction θ. Together:

Key Formula: Gabor Filter in Fourier Space

$$G(u,v) = \frac{2\pi\sigma^2}{\gamma}\, e^{i\phi} \exp\!\left(-2\pi^2\sigma^2\left[\left(\bar{u} - \frac{1}{\lambda}\right)^2 + \frac{\bar{v}^2}{\gamma^2}\right]\right), \qquad \begin{aligned}\bar{u} &= u\cos\theta + v\sin\theta \\ \bar{v} &= -u\sin\theta + v\cos\theta\end{aligned} \tag{0.47}$$

In Fourier space, a Gabor filter is a single Gaussian blob centered at distance $1/\lambda$ from the origin in direction θ. Its width is $1/(2\pi\sigma)$ along that direction and $\gamma/(2\pi\sigma)$ across it: the wider the filter in image space, the narrower its blob in Fourier space.

Example

A filter with $\lambda = 8$, $\theta = 30°$, $\sigma = 4$, and $\gamma = 0.5$ should have its blob at $u = \cos 30°/8 = 0.1083$ and $v = \sin 30°/8 = 0.0625$ cycles per pixel, with widths $1/(2\pi \cdot 4) = 0.0398$ along the direction θ and $0.5/(2\pi \cdot 4) = 0.0199$ across it. The FFT of the sampled kernel on a 256 by 256 grid has its maximum at $u = 0.1094$, $v = 0.0625$, the frequency bin closest to the prediction (bins are $1/256 = 0.0039$ apart). The measured widths of the blob are 0.0398 and 0.0199, as predicted. A real-valued kernel that keeps only the cosine part of the filter has two blobs, at $\pm(0.1094, 0.0625)$, because the cosine is the sum of two complex exponentials with opposite frequencies.

The formula explains the layout of a filter bank in Fourier space, shown on the right of Figure %s. Each direction places its blobs along one ray from the center, and each scale places them on one ring. Directions only need to cover half a circle, $\theta_k = k\pi/K$ for $k = 0, \ldots, K-1$. For phase $\phi = 0$, the filter for $\theta + \pi$ is the complex conjugate of the filter for θ, and the phase only rotates the complex response as a whole. On a real image, both directions therefore give responses of the same magnitude. Scales are usually spaced by a factor of two in frequency, with σ growing in proportion to λ so that every blob covers the same relative band. With $\lambda = 4, 8, 16$ and $\sigma = \lambda/2$, the blobs lie on rings of radius 0.25, 0.125, and 0.0625 cycles per pixel, with widths 0.080, 0.040, and 0.020 along the ray. High-frequency filters therefore appear as large blobs far from the center, and low-frequency filters as small blobs close to it. The typical frequencies of \#image-features-edges-texture-shape, between about 0.05 and 0.5 cycles per pixel, end at the Nyquist limit: a pattern that alternates every pixel is the finest one that pixels can represent. A bank with 6 directions and 3 scales, for example, has 18 filters.

Reading the Examples

A straight, high-contrast edge concentrates its energy along a line through the center of the spectrum, perpendicular to the edge. In the windsurfer photo of Figure %s, the mast against the white foam creates such a spike. The three filters whose blobs lie on this spike show the mast; the filter with a different direction misses it.

Figure 0.15 applies four filters with the same directions and scales to a mountain photo. The spectrum at the bottom left shows bright streaks through the center, produced by the ridge lines. Each filter selects a blob in this spectrum, and the inverse FFT shows where in the image the selected frequencies occur. Scale 3, the lowest frequency, responds with broad blobs in the clouds and along the horizon. Scales 2 and 1 respond to increasingly thin structures along the mountain contour. Direction 4 at scale 1 picks up different segments of the contour than direction 2, because the ridges on the two sides of the summit run in different directions.

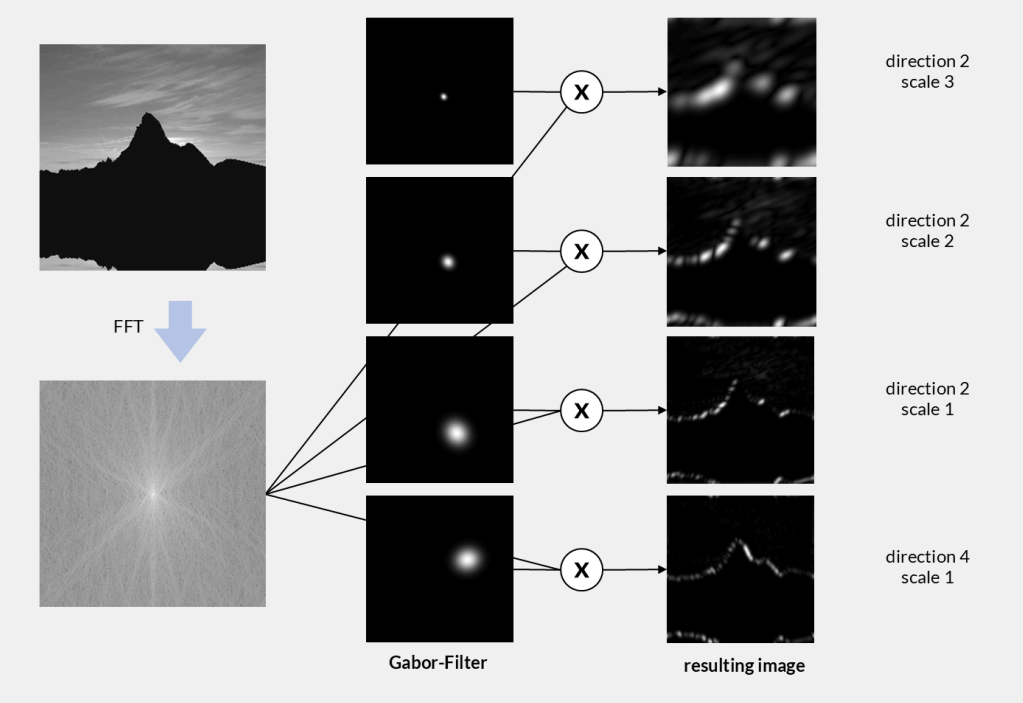


Figure 0.15: A Gabor filter bank applied in Fourier space to a mountain photo: spectrum, four filters, and the filtered images.

FFT Sizes and Padding

FFT libraries such as FFTW and the FFTs in NumPy and SciPy handle images of any size in $O(n \log n)$ time, not only sizes that are powers of two. They are fastest when the size has only small prime factors such as 2, 3, 5, and 7, and slowest for large primes. In a measurement with SciPy, a 1000 by 1000 FFT ($1000 = 2^3 \cdot 5^3$) took 10.7 ms and a 1024 by 1024 FFT 10.5 ms, while a 1009 by 1009 FFT (1009 is prime) took 56.6 ms, about five times longer. The Matterhorn photo of \#image-features has 853 rows, also a prime number; `scipy.fft.next_fast_len(853)` suggests padding to $864 = 2^5 \cdot 3^3$ rows. Padding is the right adjustment; resizing would change the image and its frequencies.

Padding matters for a second reason. The discrete Fourier transform treats the image as periodic, as if its right border continued with its left border and its bottom with its top. The product of two spectra therefore computes a circular convolution, in which a filter near the border reaches across to the opposite side. For an ordinary convolution with a $K \times K$ kernel, we pad the image with at least $K - 1$ additional rows and columns. On a 400 by 600 crop of the Matterhorn photo, filtering with a 31 by 31 Gabor kernel without padding produces errors of up to 3229 in the first 30 rows and columns, more than the largest correct response in the crop (2849). Away from these borders, the result agrees with direct convolution to within 10^{-11}, and with padding to 430 by 630 it agrees everywhere. Zero padding creates an artificial edge at the image border; padding with a mirrored copy of the image avoids it.

Spatial Convolution or FFT

A Gabor filter can be applied in two ways: directly in image space, by sliding the kernel over the image and computing a weighted sum at every pixel as in \#image-features-edges-texture-shape, or in Fourier space with the FFT pipeline above. Both are practical, and their costs grow differently. A direct convolution with a $K \times K$ kernel costs K^2 multiply-adds per pixel and filter, so its cost grows with the kernel. The FFT costs a few operations per pixel that grow only with the logarithm of the image size, independent of the kernel. On the full 853 by 1280 Matterhorn photo with a 31 by 31 kernel, SciPy's direct convolution took 2.0 seconds and its FFT-based convolution 0.06 seconds. OpenCV's `filter2D` took 0.016 seconds, because it switches to a DFT-based algorithm for kernels of about 11 by 11 pixels or larger and uses direct convolution for smaller ones. For small kernels, direct convolution is faster. In practice, we can leave the choice to the library: `filter2D` in OpenCV and `scipy.signal.convolve` with its default `method="auto"` both pick an algorithm.

The libraries differ in what they return. OpenCV's `getGaborKernel` returns a real kernel, the real part of the filter with phase `psi`. The default phase is $\pi/2$, which gives an odd, sine-shaped kernel. To obtain the magnitude of the complex response, we filter with two kernels of phase 0 and $\pi/2$ and combine the two results r_0 and $r_{\pi/2}$ as $\sqrt{r_0^2 + r_{\pi/2}^2}$. scikit-image's `skimage.filters.gabor` returns the real and imaginary responses directly; it describes the filter by its frequency and bandwidth instead of λ and σ.

From Filtered Images to Features

After filtering, we summarize each filtered image with histograms or moments. With many filters, moments are preferred: they keep the feature short and avoid the quadratic distance over very large histograms. Let $\tilde{f}_i(x, y)$ be the complex output of the i-th filter. We treat the magnitudes $\mid \tilde{f}_i(x, y) \mid$ as data points and compute their mean μ_i, standard deviation, skewness, and kurtosis, in one pass if we like. The covariance between two filters i and j multiplies the two deviations:

$$\mathrm{cov}_{i,j} = \frac{1}{NM} \sum_{x,y} \big(\mid \tilde{f}_i(x, y) \mid - \mu_i \big) \big(\mid \tilde{f}_j(x, y) \mid - \mu_j \big) \tag{0.48}$$

For a bank of 15 filters, these moments and covariances form the 165-value feature of the example in \#image-features-edges-texture-shape.

Summary

Key Takeaways

The Human Visual System

1. The light that reaches the eye is the product of illumination and reflectance, $f = l \cdot r$. Illumination varies by a factor of 100,000 and more, reflectance by about one hundred, and the visual system estimates reflectance rather than the product (lightness constancy).
2. Three cone types with peaks near 420, 534, and 564 nm provide color vision in daylight; one rod type with a peak near 498 nm provides night vision without color. The L cone peaks in the yellow-green, not in the red.
3. A retina holds on average 4.6 million cones and 92 million rods. S cones are rare (about 7 percent near the fovea), and the L:M ratio varies widely between people with normal color vision.
4. Normal acuity of 20/20 means resolving details of 1 arcminute, or 30 cycles per degree; healthy young adults see better, and the limit of the eye is about 60 cycles per degree. Eagles resolve about twice as much detail; cats, dogs, rodents, and insects resolve much less.
5. Retinal ganglion cells have center-surround receptive fields that respond to local contrast, not to brightness. Their computation is a convolution with a kernel whose weights sum to zero.
6. V1 contains simple cells, which detect oriented edges at a position, and complex cells, which tolerate shifts. This hierarchy inspired the Neocognitron and, through it, convolutional networks with convolution and pooling layers.
7. Beyond V1, the ventral stream recognizes objects and the dorsal stream processes motion and location; cortical magnification devotes most of the cortex to the fovea.

Color Systems

1. Each cone type integrates the spectrum weighted by its sensitivity, so color reduces a continuous spectrum to three numbers. Different spectra with the same three numbers look identical (metamers), which is also what lets three primaries reproduce colors.
2. CIE XYZ integrates the spectrum with the color matching functions of the 1931 standard observer. Y is the luminance, all visible colors have non-negative coordinates, and the XYZ primaries are imaginary.
3. The chromaticity (x, y) normalizes XYZ by its sum and removes brightness. It does not change when the light gets brighter or darker, which makes it useful for shadow removal.
4. sRGB uses the Rec. 709 primaries, the D65 white point, and a nonlinear transfer function with an approximate gamma of 2.2. The encoding subtracts 0.055, and matrices convert only between XYZ and linear RGB, never sRGB values directly.
5. CIE Lab compresses the ratios to the white point with a cube root to make Euclidean distances approximately perceptual. The four-box example shows equal sRGB steps of 100 as Lab distances of 42.7, 59.8, and 27.7.
6. HSL and HSV are cylindrical versions of the RGB cube. They separate hue from brightness, but they are not perceptually uniform, and their saturation values differ for the same color.

Extended Image Features

1. Raw sums of the first four powers and of the channel products yield all moments and covariances in a single pass. The formulas cancel large terms, so they need 64-bit accumulators, shifted data, or update formulas that keep central moments.
2. The quadratic distance becomes a squared Euclidean distance after the transformation $\boldsymbol{\Lambda}^{1/2}\boldsymbol{U}^\top$, provided that $\boldsymbol{A}$ is positive semidefinite. Dropping axes with small eigenvalues gives a lower bound for a filter step.
3. The matrix $a_{i,j} = 1 - d_{i,j}/\max d$ is not positive semidefinite in general. For normalized histograms, centering the matrix fixes this without changing any distance; a Gaussian kernel is positive definite by construction.
4. Gradient moments summarize magnitude and direction with nine values, but the direction is an angle and needs circular treatment.

5. Laws' texture energy derives 16 kernels from four prototype vectors and reduces them to 9 energy maps. Symmetric pairs are combined by averaging their energy maps, not their kernels. The maps give 512-bin histograms or 72 moments.
6. In Fourier space, a Gabor filter is a Gaussian blob at distance $1/\lambda$ from the center in direction θ; a filter bank covers a half circle of directions and a few rings of scales.
7. FFTs handle any image size and are fastest for sizes with small prime factors. Padding avoids the wrap-around of circular convolution. Libraries choose between direct and FFT-based convolution automatically.

Key Formulas

Key Formula: Image Formation

$$f(x, y, z) = l(x, y, z) \cdot r(x, y, z) \tag{0.49}$$

The light that a point sends toward the viewer is the product of illumination and reflectance; vision estimates the reflectance.

Key Formula: CIE XYZ Tristimulus Values

$$X = k \int_{380}^{780} f(\lambda)\, \bar{x}(\lambda)\, d\lambda, \qquad Y = k \int_{380}^{780} f(\lambda)\, \bar{y}(\lambda)\, d\lambda, \qquad Z = k \int_{380}^{780} f(\lambda)\, \bar{z}(\lambda)\, d\lambda \tag{0.50}$$

Each tristimulus value weights the spectrum with one color matching function of the 1931 standard observer.

Key Formula: CIE Lab

$$L^* = 116 \cdot \left(f\frac{Y}{Y_n}\right) - 16, \qquad a^* = 500 \cdot \left(\left(f\frac{X}{X_n}\right) - \left(f\frac{Y}{Y_n}\right)\right), \qquad b^* = 200 \cdot \left(\left(f\frac{Y}{Y_n}\right) - \left(f\frac{Z}{Z_n}\right)\right) \tag{0.51}$$

With $f(t) = \sqrt[3]{t}$ above a small threshold, the cube roots of the ratios to the white point make Euclidean distances approximately perceptual.

Key Formula: Quadratic Distance as a Euclidean Distance

$$\delta_{\text{quad}}(\boldsymbol{h}, \boldsymbol{g}) = \| \, \boldsymbol{\Lambda}^{1/2} \boldsymbol{U}^\top (\boldsymbol{h} - \boldsymbol{g}) \, \|^2 \tag{0.52}$$

For a positive semidefinite $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$, transforming every histogram once turns the quadratic distance into a squared Euclidean distance.

Further Reading

The Human Visual System

- Hubel, D. H., & Wiesel, T. N. (1962). **Receptive fields, binocular interaction and functional architecture in the cat's visual cortex**. *The Journal of Physiology*, 160(1), 106-154. PubMed Central. The classic description of simple and complex cells in V1, the starting point of the hierarchy that convolutional networks imitate.
- Fukushima, K. (1980). **Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position**. *Biological Cybernetics*, 36(4), 193-202. PDF. Turns the simple and complex cells into a layered network of S-cells and C-cells, the direct ancestor of convolution and pooling layers.
- Curcio, C. A., Sloan, K. R., Kalina, R. E., & Hendrickson, A. E. (1990). **Human photoreceptor topography**. *Journal of Comparative Neurology*, 292(4), 497-523. PubMed. The source of the rod and cone counts and the foveal cone density used in E.1.
- Curcio, C. A., Allen, K. A., Sloan, K. R., Lerea, C. L., Hurley, J. B., Klock, I. B., & Milam, A. H. (1991). **Distribution and morphology of human cone photoreceptors stained with anti-blue opsin**.

Journal of Comparative Neurology, 312, 610-624. PubMed. Shows that S cones make up about 7 percent of the cones and are missing at the very center of the fovea.
- Curcio, C. A., & Allen, K. A. (1990). **Topography of ganglion cells in human retina**. *Journal of Comparative Neurology*, 300(1), 5-25. PubMed. Counts 0.7 to 1.5 million ganglion cells per retina and links cortical magnification to ganglion cell density.
- Hofer, H., Carroll, J., Neitz, J., Neitz, M., & Williams, D. R. (2005). **Organization of the human trichromatic cone mosaic**. *The Journal of Neuroscience*, 25(42), 9669-9679. PubMed Central. Images of the cone mosaic in living eyes, with L:M ratios from 1.1:1 to 16.5:1 among people with normal color vision.
- Goodale, M. A., & Milner, A. D. (1992). **Separate visual pathways for perception and action**. *Trends in Neurosciences*, 15(1), 20-25. PubMed. The proposal of the ventral stream for recognition and the dorsal stream for visually guided action.
- Caves, E. M., Brandley, N. C., & Johnsen, S. (2018). **Visual acuity and the evolution of signals**. *Trends in Ecology & Evolution*, 33(5), 358-372. DOI. A survey of acuity across animals in cycles per degree, showing that most species see far less detail than humans.
- Reymond, L. (1985). **Spatial visual acuity of the eagle Aquila audax: A behavioural, optical and anatomical investigation**. *Vision Research*, 25(10), 1477-1491. PubMed. The measurement behind the eagle's acuity of about 140 cycles per degree, twice the human limit.
- Allison, W. T., Haimberger, T. J., Hawryshyn, C. W., & Temple, S. E. (2004). **Visual pigment composition in zebrafish: Evidence for a rhodopsin-porphyropsin interchange system**. *Visual Neuroscience*, 21(6), 945-952. PubMed. Microspectrophotometry of zebrafish photoreceptors, the source of the four measured cone peaks from 361 to 565 nm.
- Kolb, H., et al. **Webvision: The organization of the retina and visual system**. Online textbook. webvision.pitt.edu. A free, detailed reference on the cell types and circuits of the retina, including bipolar, ganglion, horizontal, and amacrine cells.

Color Systems

- Smith, T., & Guild, J. (1931). **The C.I.E. colorimetric standards and their use**. *Transactions of the Optical Society*, 33(3), 73-134. IOP Science. The original description of the 1931 standard observer and of the XYZ system that every later CIE space builds on.
- CIE (2018). **Colorimetry** (4th ed., CIE 015:2018). International Commission on Illumination. CIE. The current standard for XYZ, xyY, and Lab, including the official tables; not free of charge.
- Colour & Vision Research Laboratory. **CVRL database: Colour matching functions**. University College London. cvrl.ucl.ac.uk. Free tables of the CIE 1931 color matching functions in 1 nm steps, the data behind the metamer example in E.2.
- Stokes, M., Anderson, M., Chandrasekar, S., & Motta, R. (1996). **A standard default color space for the Internet: sRGB** (Version 1.10). W3C. w3.org. The original proposal by HP and Microsoft that became IEC 61966-2-1, with the primaries and the transfer function.
- ITU-R (2015). **Recommendation BT.2020-2: Parameter values for ultra-high definition television systems for production and international programme exchange**. International Telecommunication Union. ITU. Defines the wide-gamut Rec. 2020 primaries used in E.2.
- Smith, A. R. (1978). **Color gamut transform pairs**. *Computer Graphics (SIGGRAPH '78 Proceedings)*, 12(3), 12-19. PDF. Introduces the HSV hexcone model and the fast conversion between RGB and HSV.
- Sharma, G., Wu, W., & Dalal, E. N. (2005). **The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations**. *Color Research & Application*, 30(1), 21-30. PDF. The reference implementation of CIEDE2000, the corrected successor of the Euclidean distance in Lab.

Extended Image Features

- Laws, K. I. (1980). **Rapid texture identification**. *Proceedings of SPIE*, 238, 376-381. SPIE Digital Library. Describes texture energy measures computed with small convolution masks, the basis of the method in E.3.
- Stricker, M. A., & Orengo, M. (1995). **Similarity of color images**. *Storage and Retrieval for Image and Video Databases III, Proceedings of SPIE*, 2420. SPIE Digital Library. Introduces color moments as a compact alternative to color histograms, the features whose one-pass computation E.3 derives.
- Hafner, J., Sawhney, H. S., Equitz, W., Flickner, M., & Niblack, W. (1995). **Efficient color histogram indexing for quadratic form distance functions**. *IEEE Transactions on Pattern Analysis and*

Machine Intelligence, 17(7), 729-736. IEEE Xplore. Shows how cheap lower bounds of the quadratic histogram distance prefilter a large image collection before the exact distance is computed.

- Manjunath, B. S., & Ma, W. Y. (1996). **Texture features for browsing and retrieval of image data**. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 18(8), 837-842. IEEE Xplore. The standard reference for Gabor texture features in image retrieval, with a comparison against other multiresolution texture features.
- Frigo, M., & Johnson, S. G. (2005). **The design and implementation of FFTW3**. *Proceedings of the IEEE*, 93(2), 216-231. PDF. Explains how a modern FFT library computes transforms of any size, including primes, in $O(n \log n)$ time.
- Pébay, P. P. (2008). **Formulas for robust, one-pass parallel computation of covariances and arbitrary-order statistical moments** (Report SAND2008-6212). Sandia National Laboratories. OSTI. Update formulas for central moments that avoid the cancellation of the raw-sum approach and merge results computed in parallel.